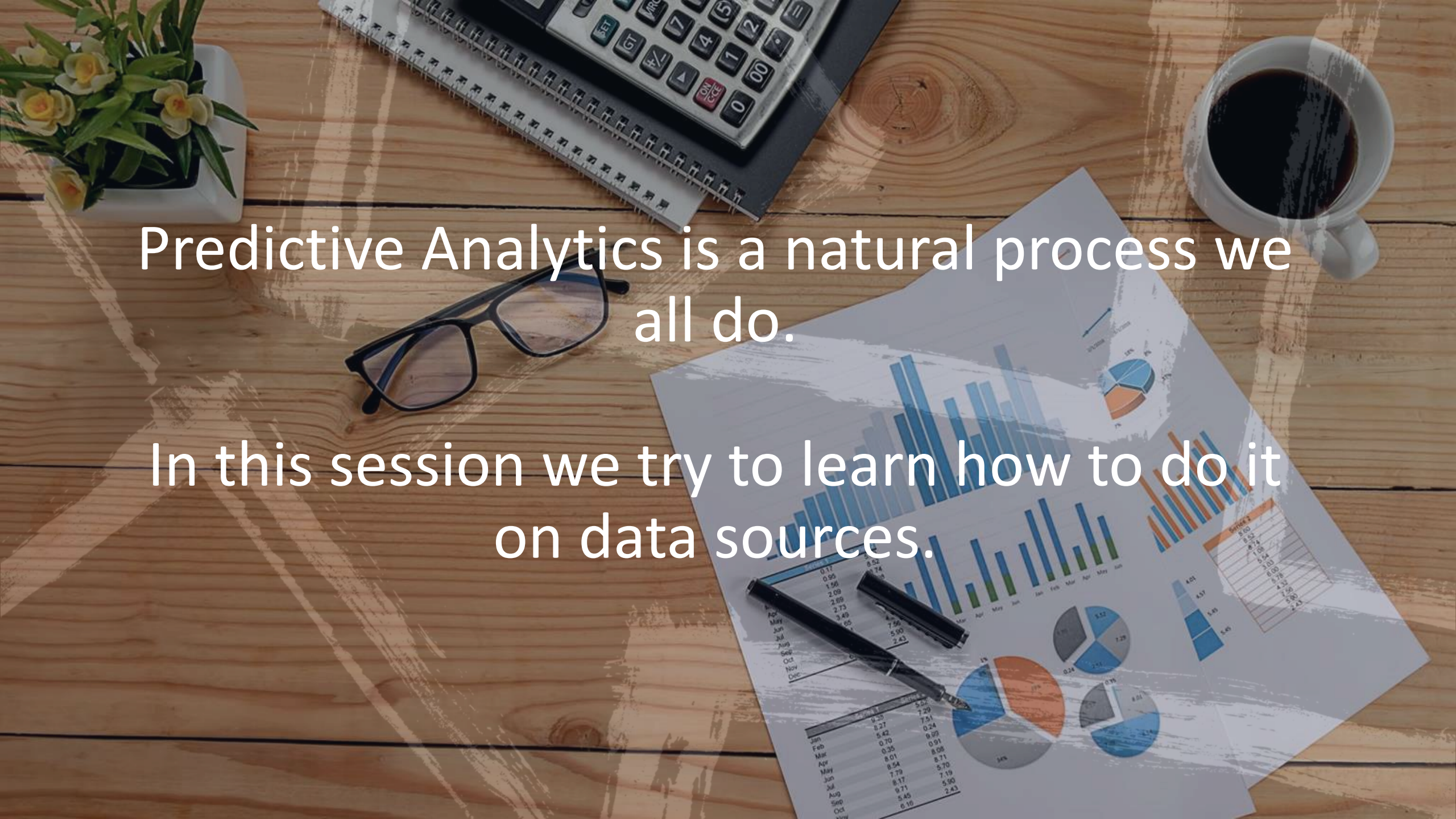


Week 7: Forecasting (Predictive Analytics) in Supply Chains (SCs)

Presented by:

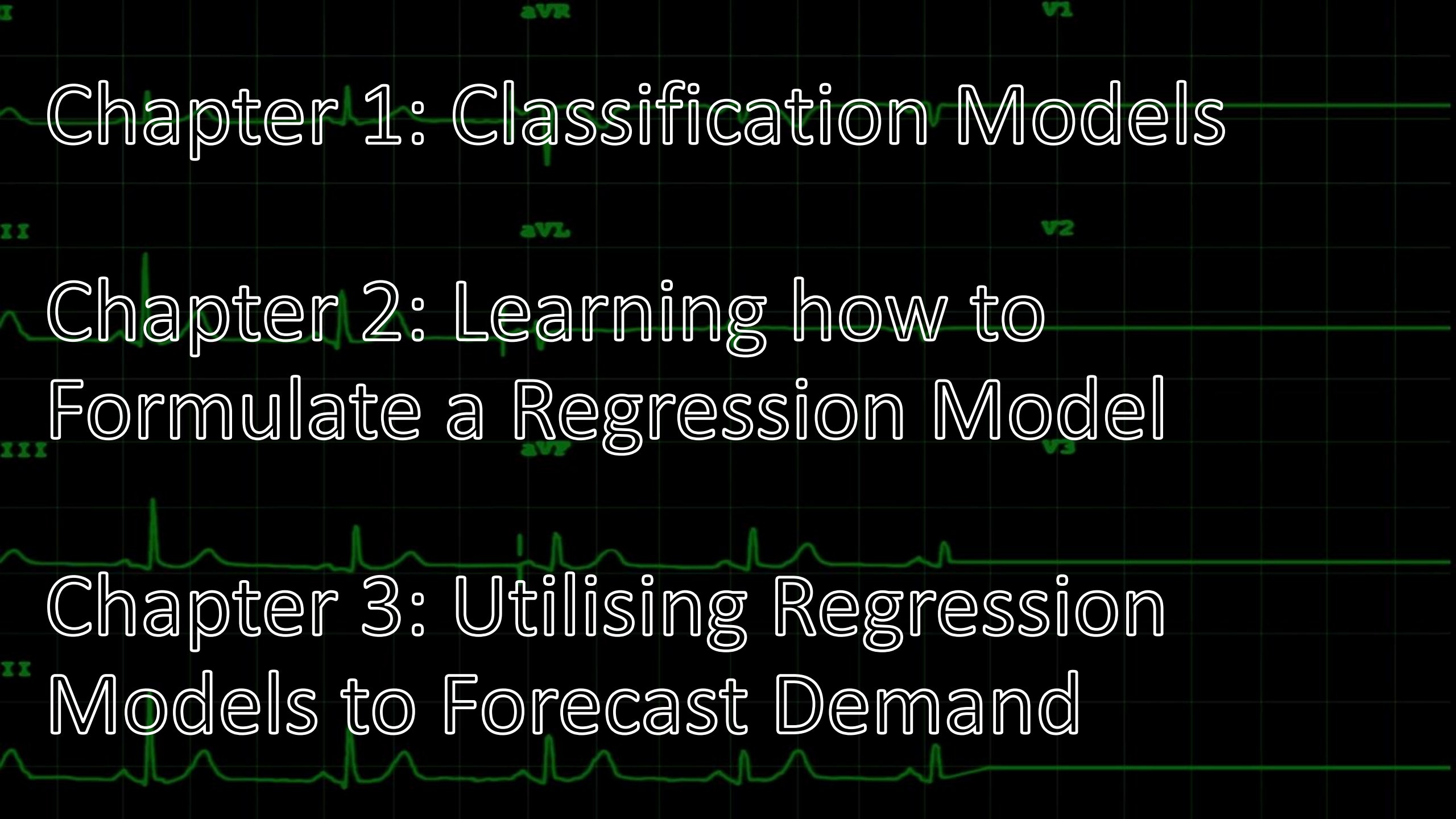
Dr. Mehdi Rajabi Asadabadi



A top-down view of a wooden desk. In the top left is a small white pot with yellow and white flowers. Next to it is a silver calculator with a numeric keypad and function buttons. To the right is a white cup filled with dark coffee. In the center-left, a pair of black-rimmed glasses lies on the desk. A large, semi-transparent white text box is centered over the desk, containing two lines of text. In the bottom right, a printed document with various charts and tables is visible, with a black pen resting on it.

Predictive Analytics is a natural process we
all do.

In this session we try to learn how to do it
on data sources.

The background of the slide features a dark green grid with a white ECG (heart rate) line. Faint green text labels for ECG leads are visible: 'I', 'aVR', 'V1' in the top row; 'II', 'aVL', 'V2' in the second row; 'III', 'aVF', 'V3' in the third row; and 'II' in the fourth row. The text is white and outlined.

Chapter 1: Classification Models

Chapter 2: Learning how to
Formulate a Regression Model

Chapter 3: Utilising Regression
Models to Forecast Demand

Chapter 1: Classification Models

Chapter 2: Learning how to Formulate a Regression Model

Chapter 3: Utilising Regression Models to Forecast Demand

Predictive Analytics



Predictive analytics focus on **using historical data** to identify patterns enabling the **prediction of the future**.



The identified pattern or trend from historical data is represented by a **mathematical model**.



This model can then be used to predict future events based on the previous data and the new data.



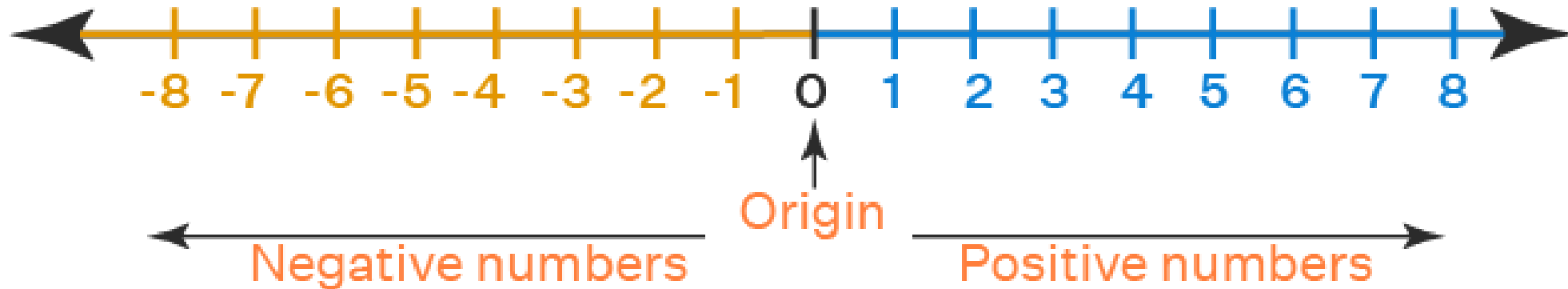
Predictive Analytics Models

Classification:

Classification is the process of creating a set of classes for data, based on the existing data.

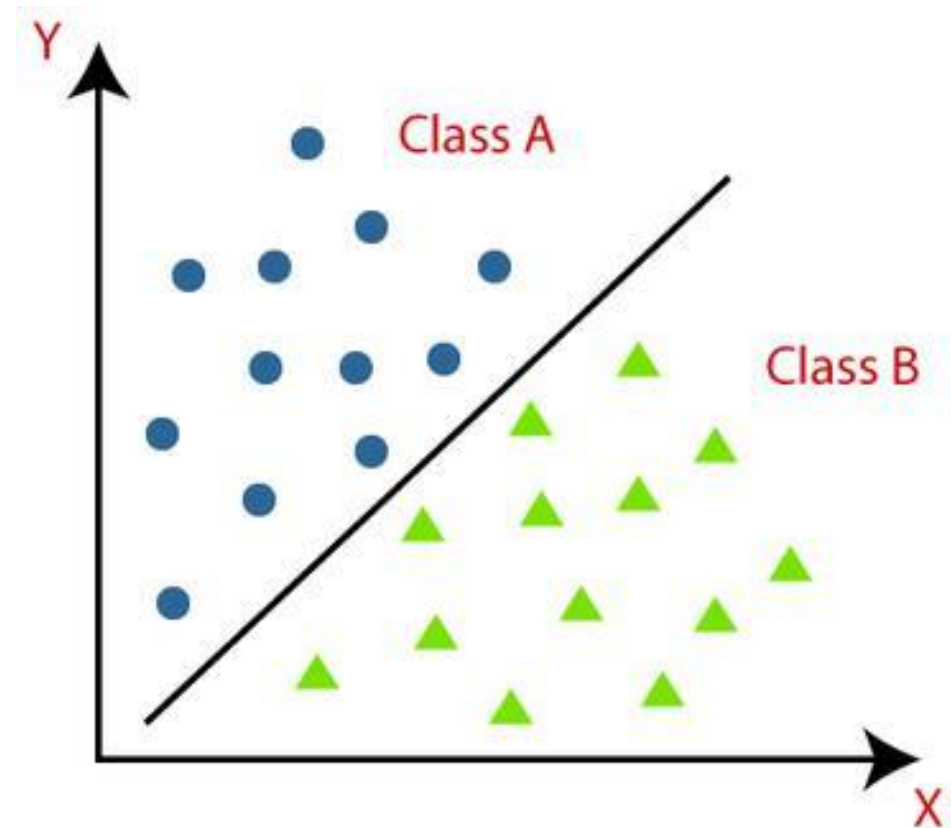
Predictive Analytics Models

Classification



Predictive Analytics Models

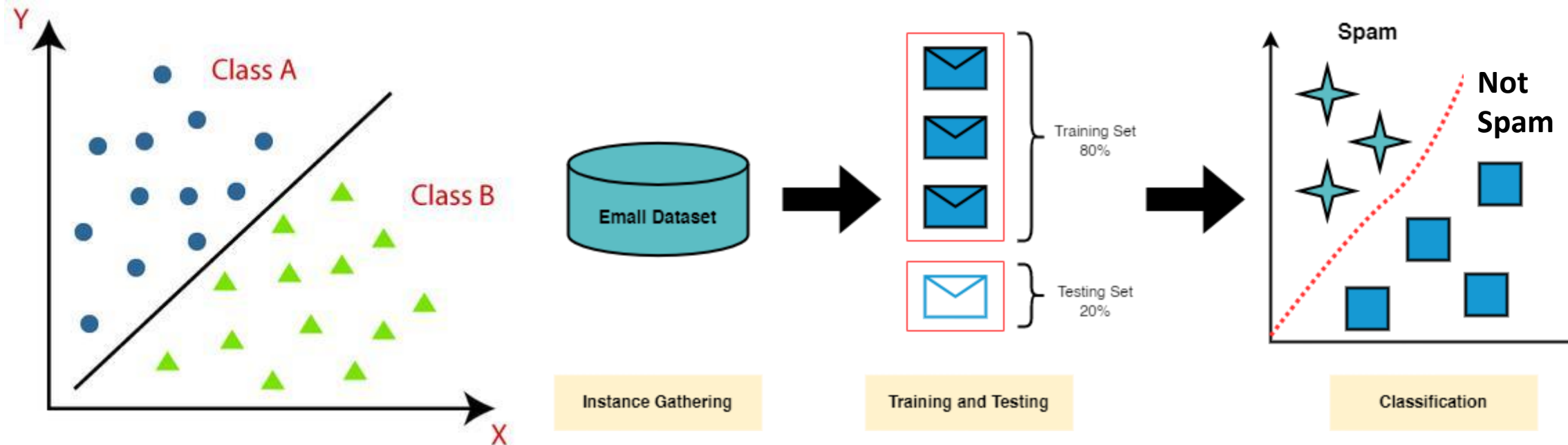
Classification



Predictive Analytics Models

Classification

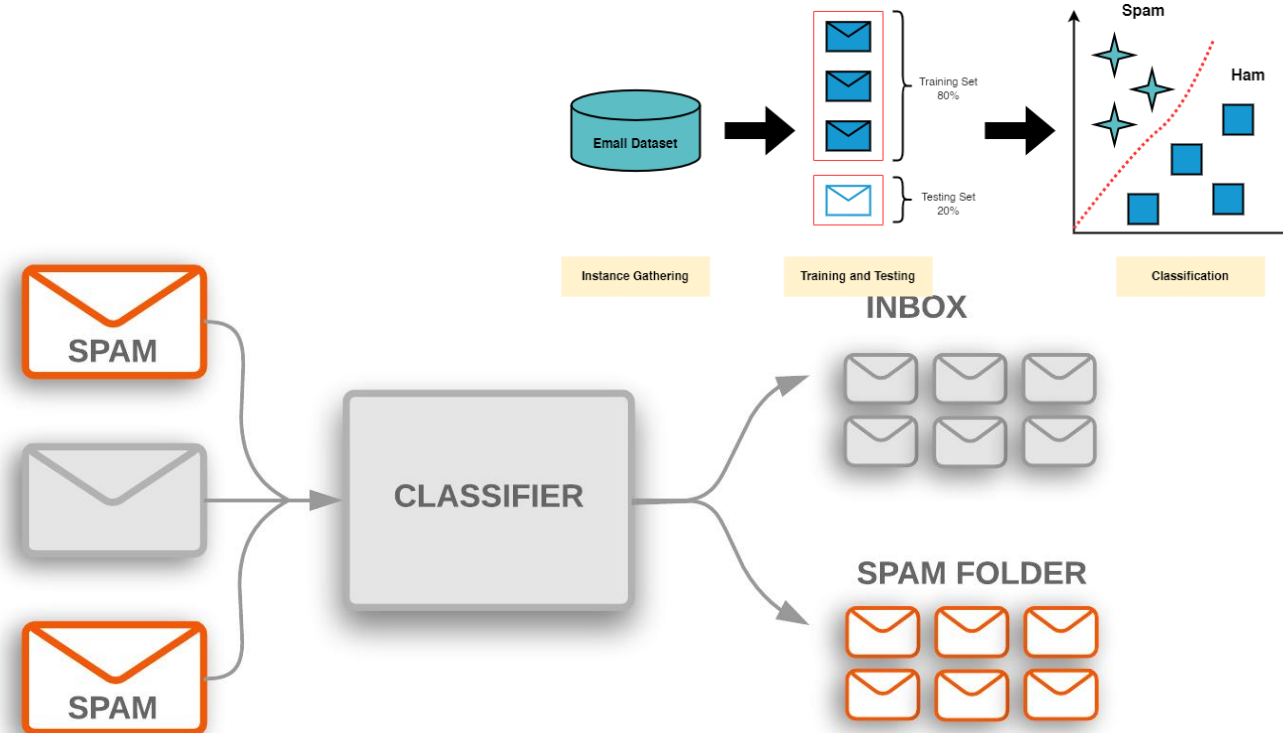
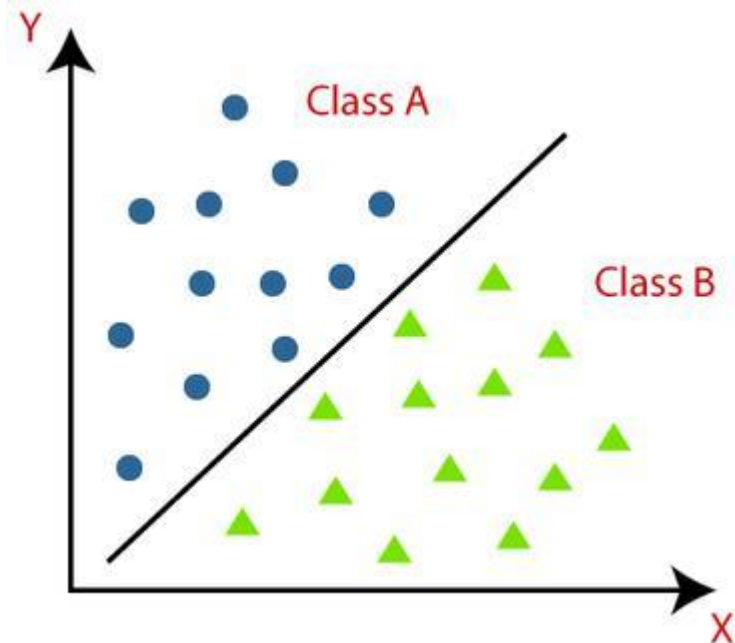
Binary classification:



Predictive Analytics Models

Classification

Binary classification:

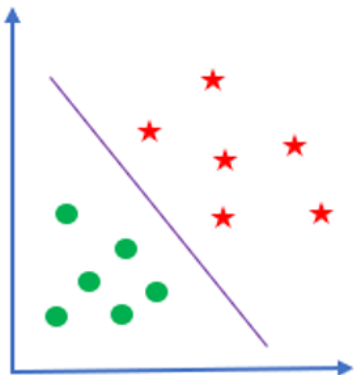


Predictive Analytics Models

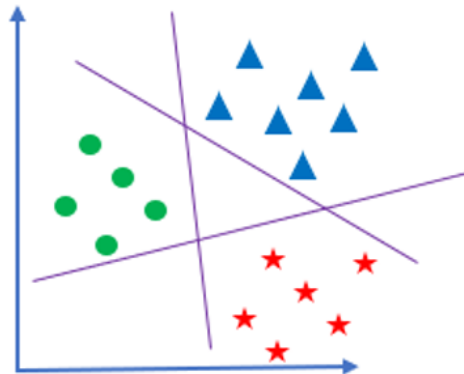
Classification

Multiclass classification:

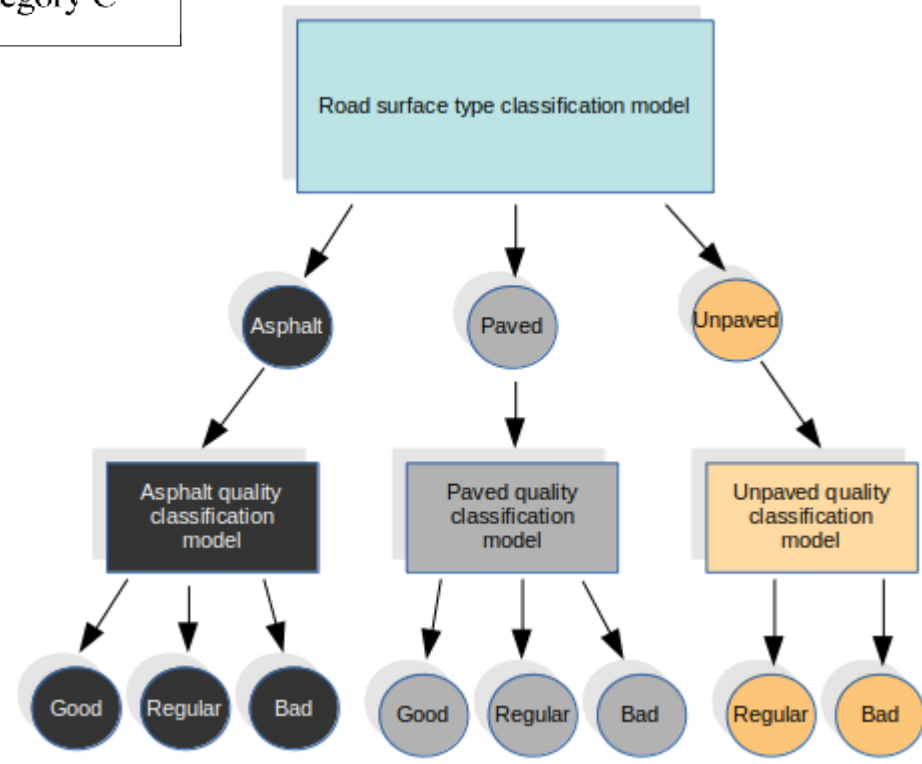
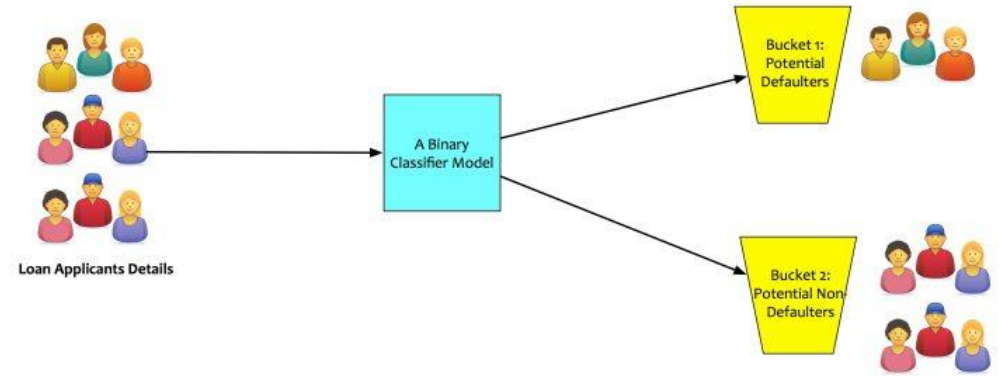
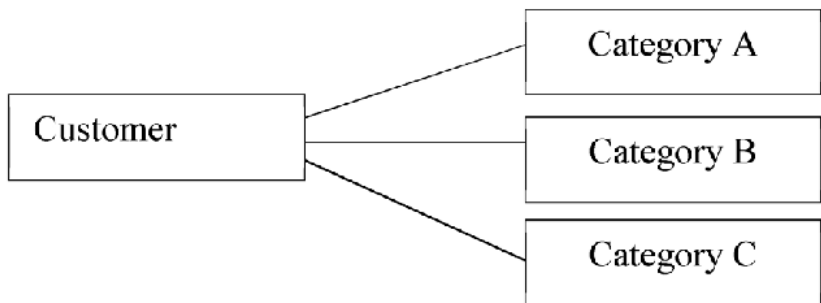
Binary classification



Multi-class classification



		SEVERITY →		
		1	2	3
LIKELIHOOD ↓	1	LOW - 1 -	LOW - 2 -	MEDIUM - 3 -
	2	LOW - 2 -	MEDIUM - 4 -	HIGH - 6 -
	3	MEDIUM - 3 -	HIGH - 6 -	HIGH - 9 -



Predictive Analytics Models

Classification:

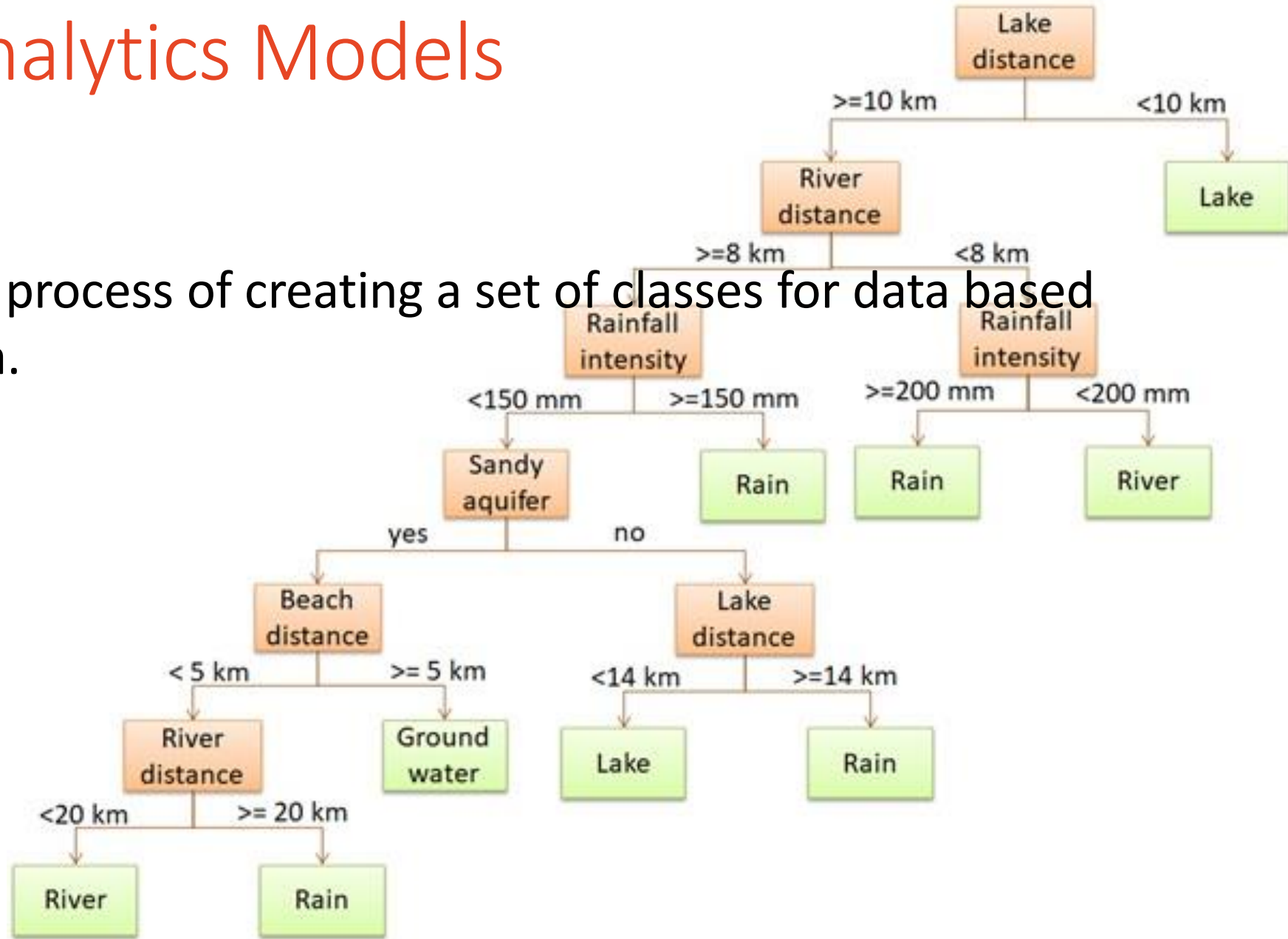
Classification is the process of creating a set of classes for data based on the existing data.

Predictive Analytics Models

Classification:

Classification is the process of creating a set of classes for data based on the existing data.

Decision trees



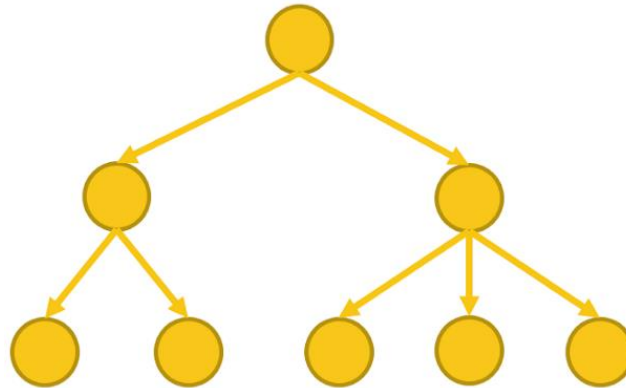
Predictive Analytics Models

Classification:

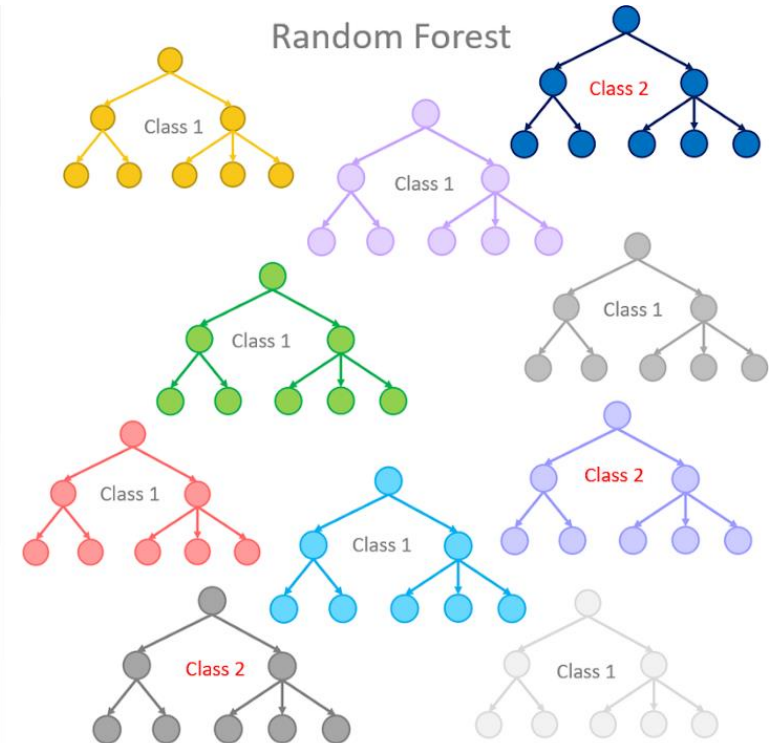
Classification is the process of creating a set of classes for data based on the existing data.

Decision trees
Random forest

Single Decision Tree



Random Forest



Predictive Analytics Models

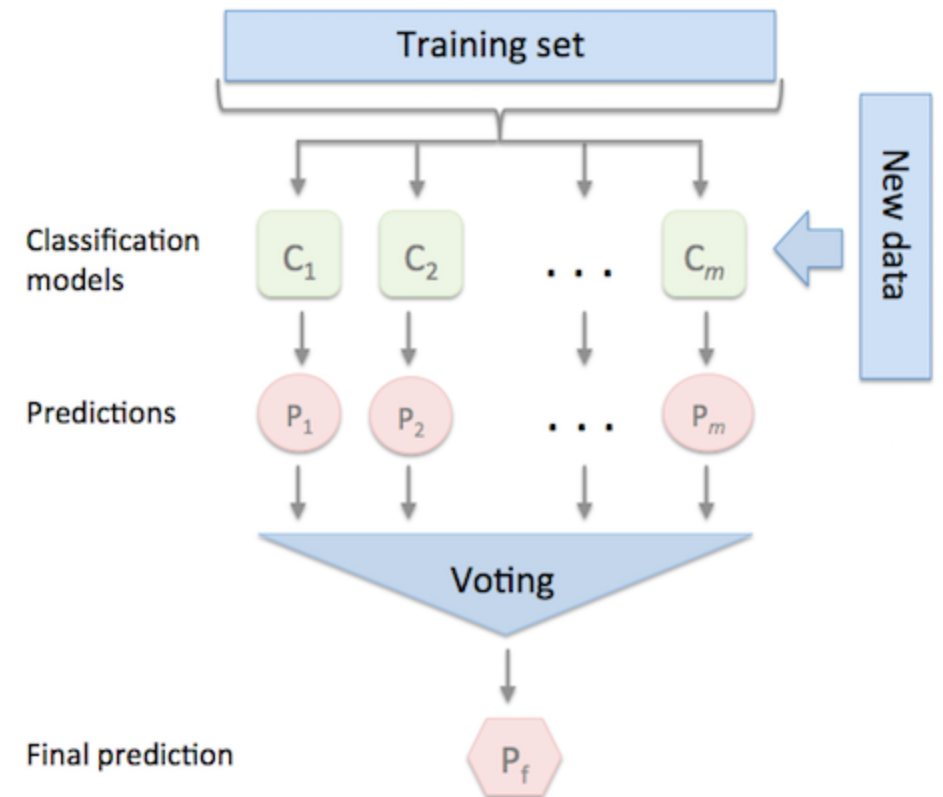
Classification:

Classification is the process of creating a set of classes for data based on the existing data.

Decision trees

Random forest

Voting classifiers



Predictive Analytics Models

Classification:

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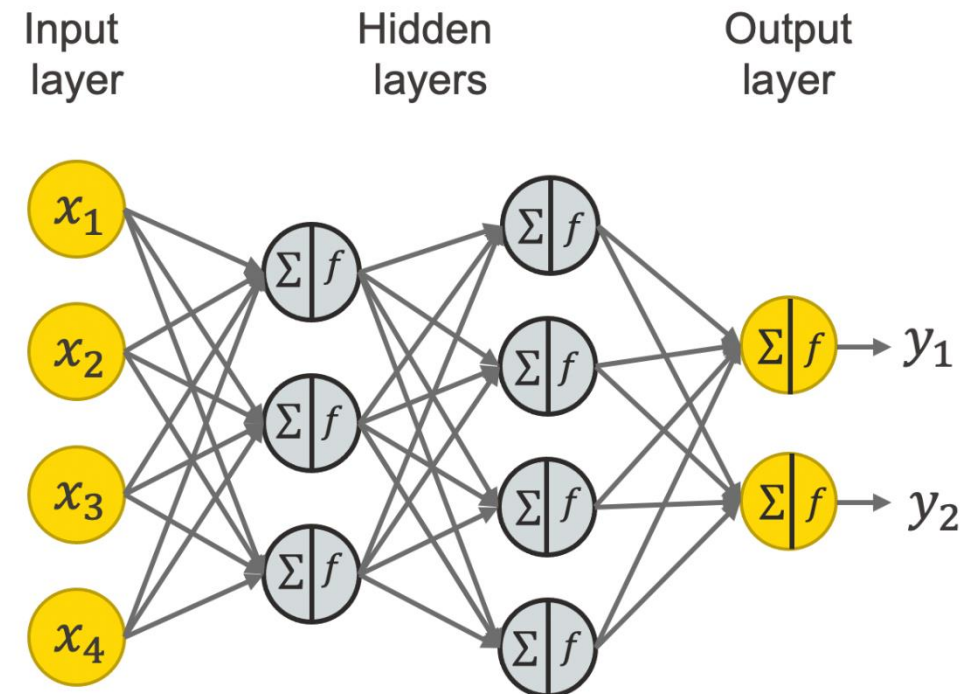
Decision trees

Random forest

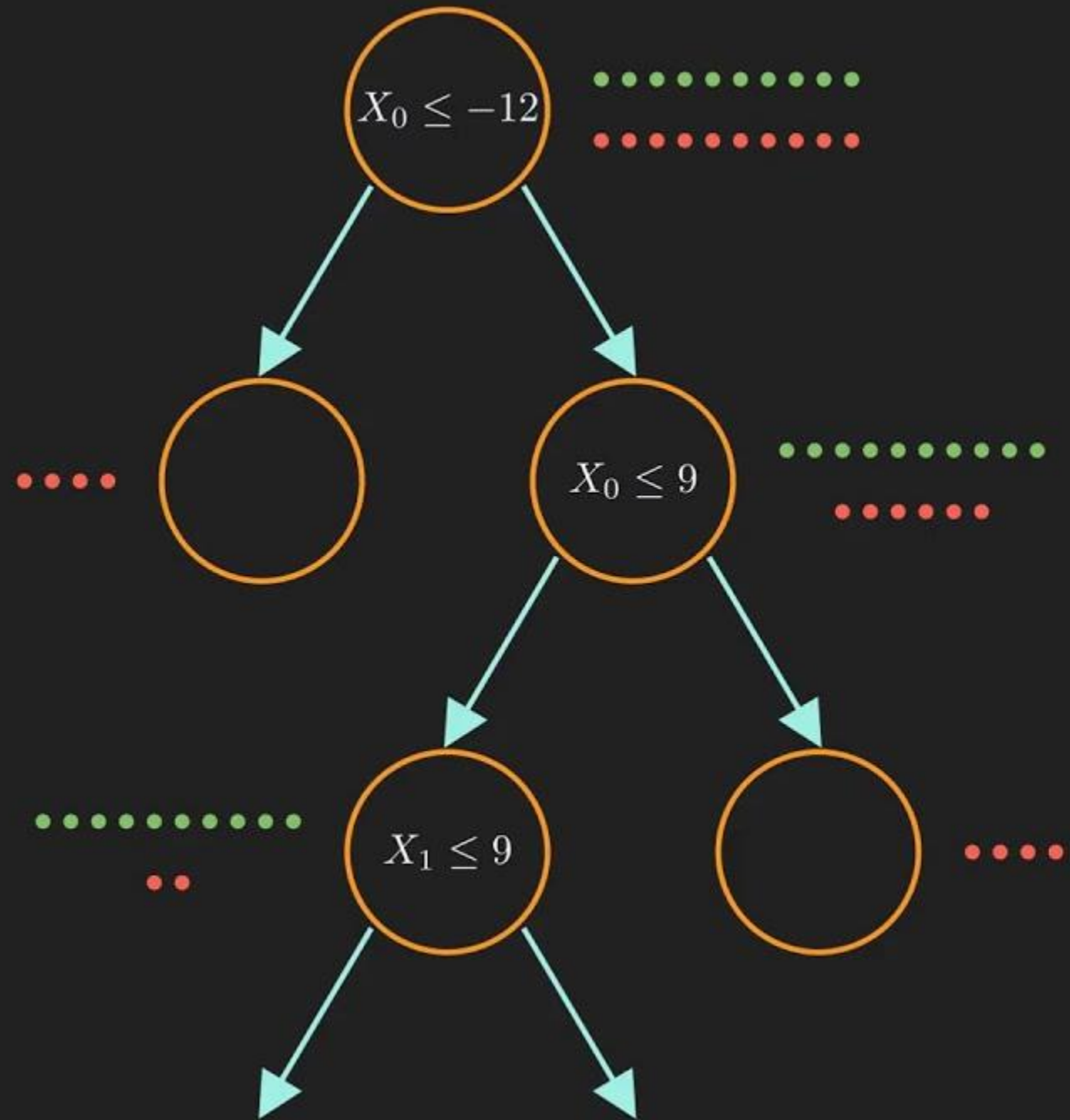
Voting classifiers

Neural networks and deep learning

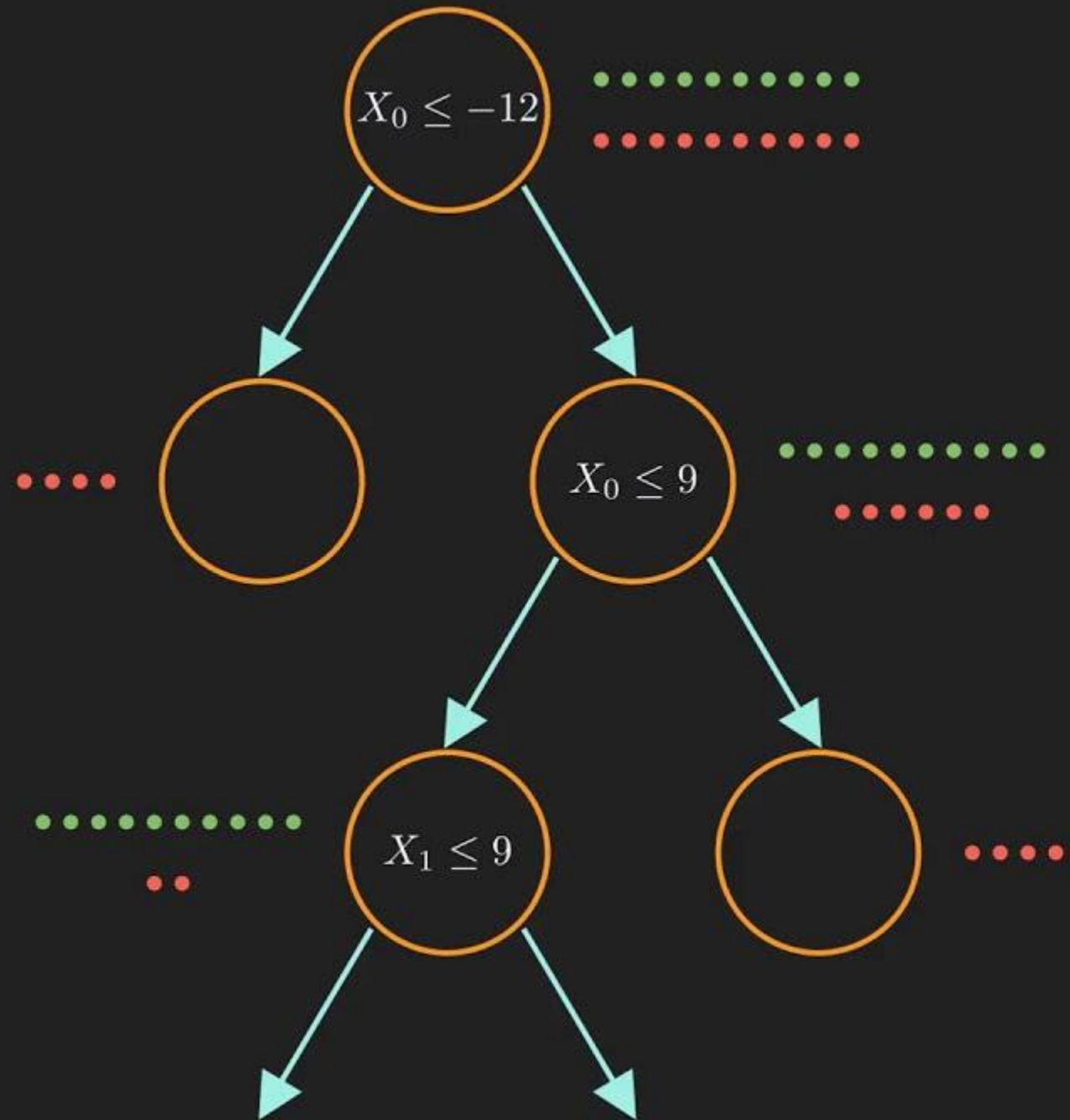
...



Decision Tree Classifier



- A decision tree is essentially an **upside-down tree shaped diagram used to classify**.
- It is a predictive model **based on a branching series of Boolean tests (often) and non-Boolean tests**.
- It has a **root node** which is the **starting point of the decision tree**.
- **Splitting or branching** is the process of dividing a node into **two or more sub-nodes**.
- Nodes have **sub-nodes and leaf/terminal nodes**, which are the ones without a split.
- **Sub-nodes** of a specific node is known as **child nodes** and the node is known as **parent node**.



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What is the first
thing we need to
do in Data
Analytics?

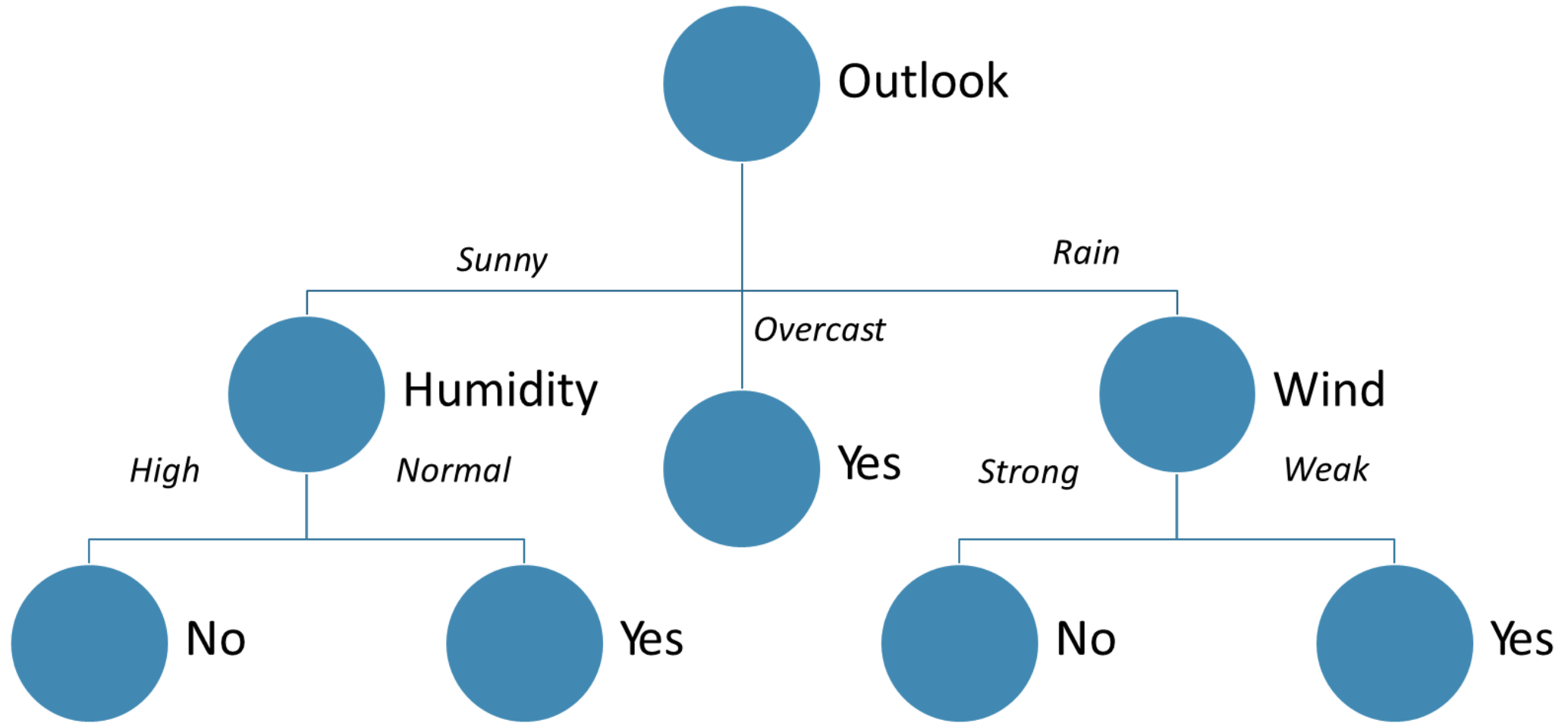
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features or attributes				decision
Outlook	Temp	Humidity	Windy	Play Tennis?
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Historical Data

features or attributes				decision/classes
Outlook	Temp	Humidity	Windy	Play Tennis?
Sunny	Hot	High	Weak	No
Sunny	Hot	High	Strong	No
Overcast	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	Cool	Normal	Strong	No
Overcast	Cool	Normal	Strong	Yes
Sunny	Mild	High	Weak	No
Sunny	Cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
Sunny	Mild	Normal	Strong	Yes
Overcast	Mild	High	Strong	Yes
Overcast	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

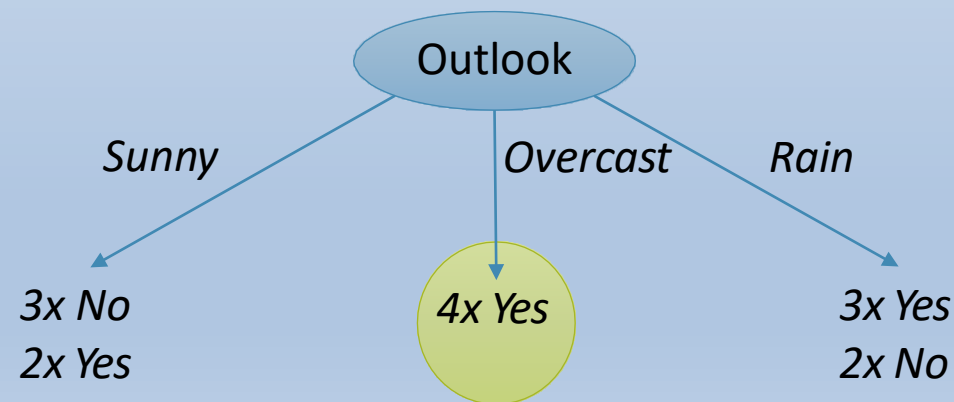
Decision Tree Classification



Decision Tree Classification

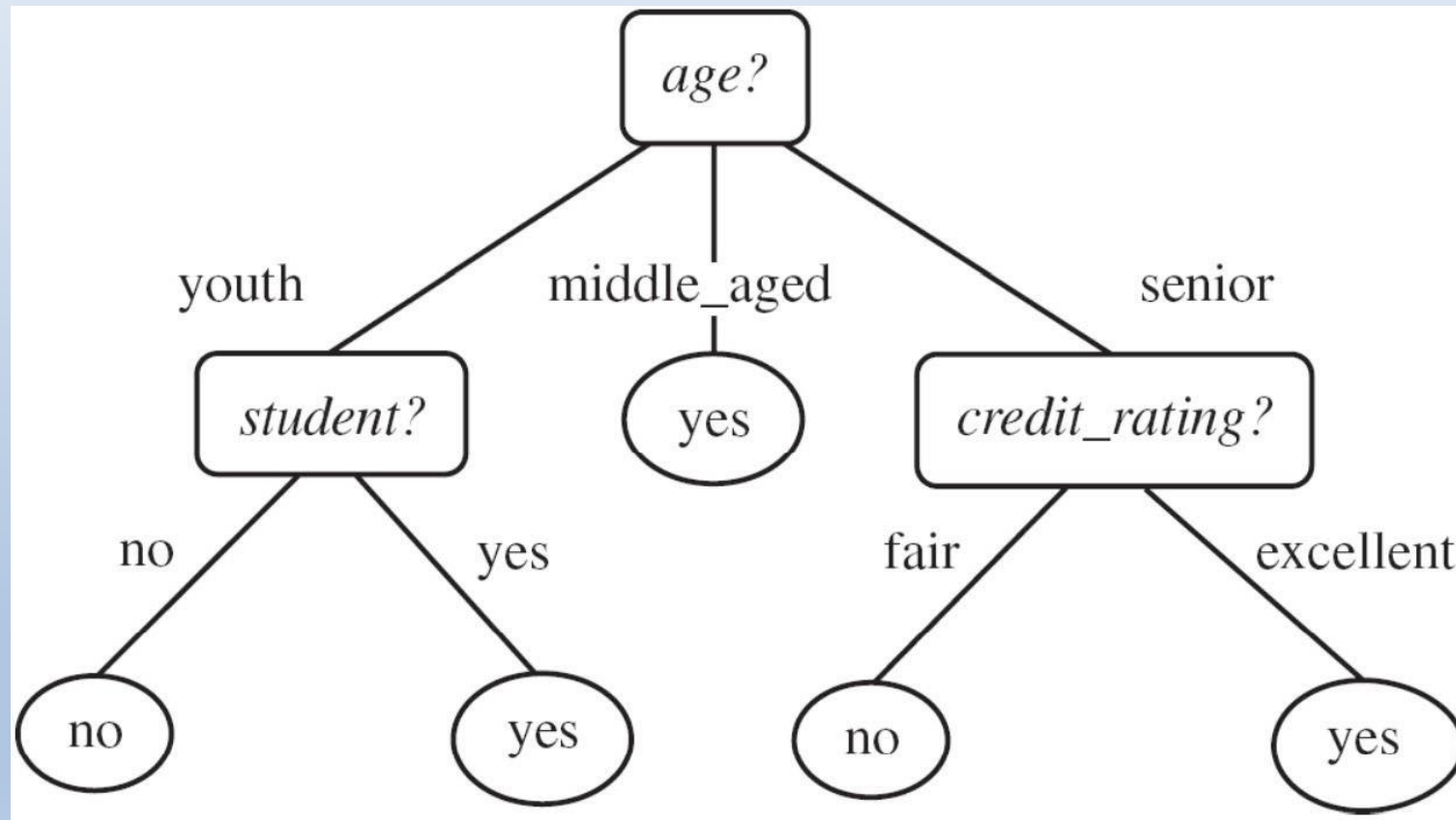
How do you build a decision tree?

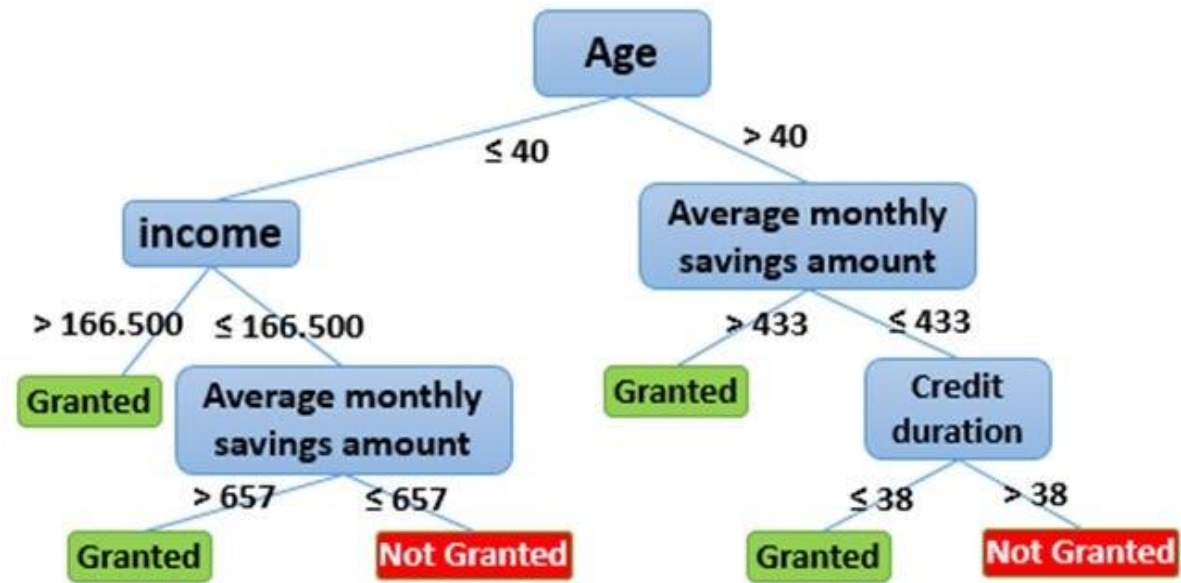
- Select an attribute (A)
 - Which attribute is the best? Impurity or information gain can be used
- For each value of A, create a partition.
- If training samples are perfectly classified, then stop – otherwise recursively iterate over the new child nodes



Example: Classifying potential customers using a decision tree

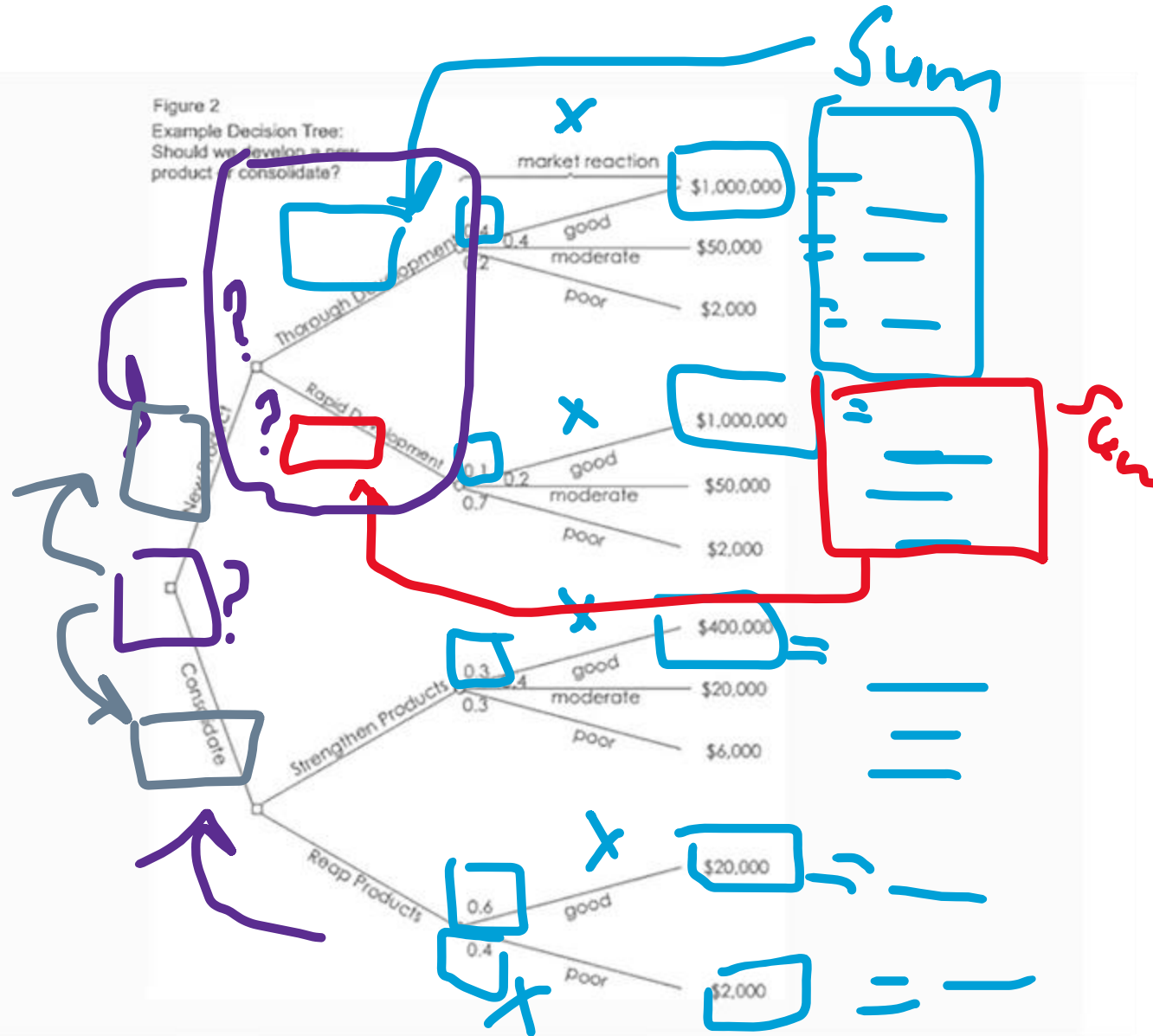
<i>RID</i>	<i>age</i>	<i>income</i>	<i>student</i>	<i>credit_rating</i>	<i>Class: buys_computer</i>
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no





Home Loan Approval

Figure 2
Example Decision Tree:
Should we develop a new
product or consolidate?





With limited attributes, it is possible to do the graphing of decision tree by hand, in other cases, it may become really time consuming and confusing.

Later in your jobs, if you wanted to use decision tree, remember that there are automated tools to do decision tree for you.

Applications of Classification Techniques

- **Financial markets** – ex: stock price prediction
- **Marketing** – next best offer prediction
- **Web Analytics** – ‘customer like you also purchased x,y,z’
- **Credit Modelling** – will you get approved for a credit card?
- **Medicine** – predicting the likely protein to bind to a virus
- **Social/Political Science** – predicting who wins the next US election
- **Automotive Industry** – Autopilot

Predictive Analytics Models in Supply Chains

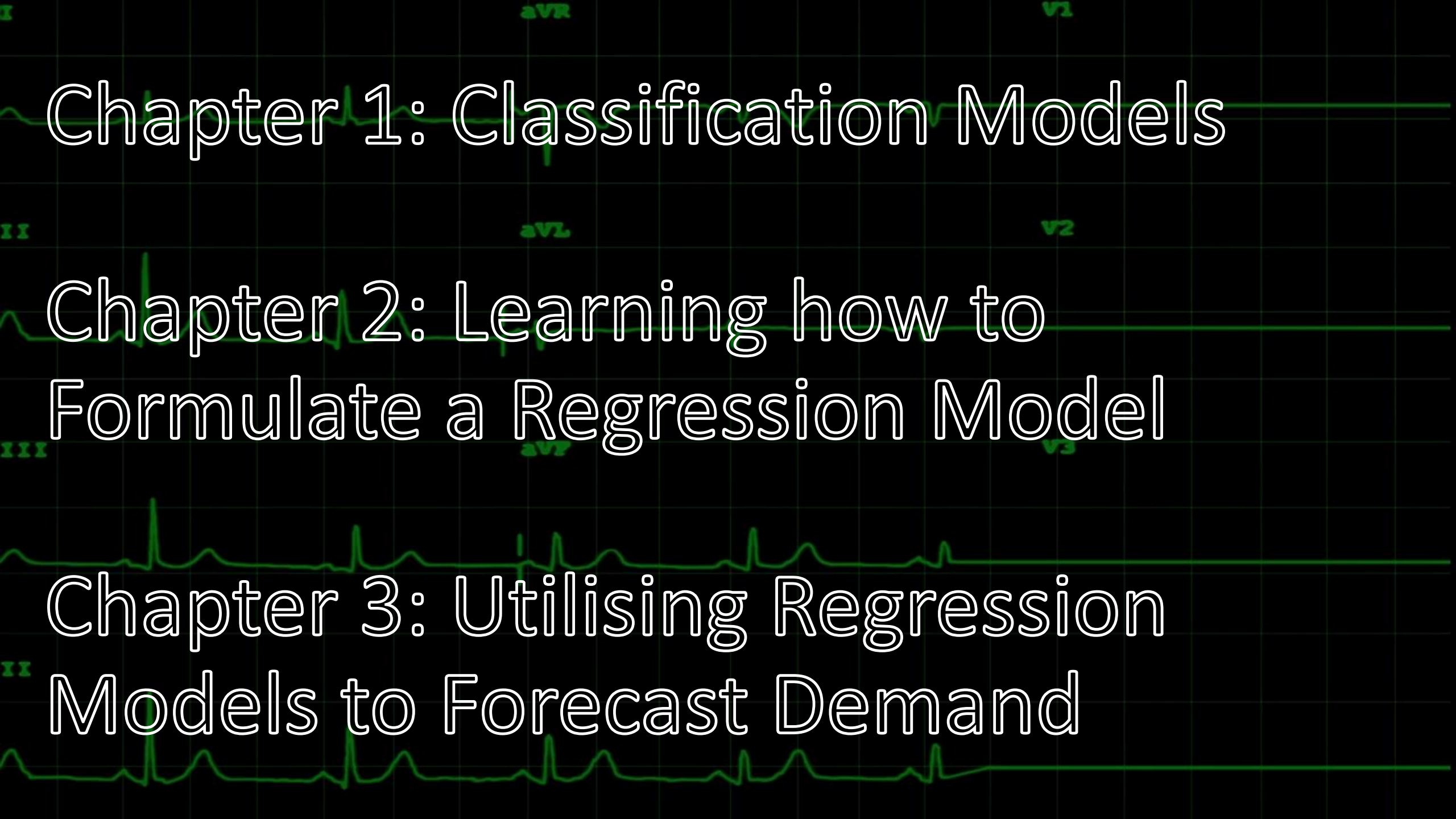
- **Classification**

Classification is a mathematical model that can differentiate **between two or more outcomes**. For instance, using the historical data of a supplier, and setting some decision attributes, the decision of ‘**whether to sign a contract with this supplier?**’ can be answered by a classification model. Managers can be provided with such classification models to assist their decisions.

- **Regression**

As opposed to predicting a decision, regression focus on **predicting an unknown future value using available data**. For instance, **given the bitcoin prices for the last year, ‘what will be the price next week?’** can be answered by a regression model. Managers can be provided with the predicted value of these models to assist their decisions.

- **Other models (Clustering, Time Series, Forecasting, and similar)**

The background of the slide features a dark green grid with a white ECG line. Faint green text labels for ECG leads are visible: 'I', 'aVR', 'V1' at the top; 'II', 'aVL', 'V2' in the middle; and 'III', 'aVF', 'V3' at the bottom.

Chapter 1: Classification Models

Chapter 2: Learning how to
Formulate a Regression Model

Chapter 3: Utilising Regression
Models to Forecast Demand

The background of the slide is a dark green grid with a white ECG (heart rate) line. The ECG line is composed of several distinct waveforms, each labeled with a lead name in green text: 'aVR' at the top, 'aVL' in the middle, and 'aVF' at the bottom. The text 'Chapter 1: Classification Models' is written in a white, sans-serif font with a thin black outline, positioned in the upper left quadrant of the slide.

Chapter 1: Classification Models

Chapter 2: Learning how to Formulate a Regression Model

Chapter 3: Utilising Regression Models to Forecast Demand

Demand forecasting is the process of predicting how much of a product or service customers will want in the future.

A photograph of a Woolworths supermarket entrance. The store's name is displayed in large, white, 3D block letters above the glass entrance. To the right of the name is a large green logo consisting of three interlocking circles. The interior of the store is visible through the glass, showing aisles with various products, including fresh produce on the left and packaged goods on the right. A person in a light-colored dress is walking away from the camera into the store. The overall lighting is bright, typical of a modern retail environment.

Woolworths



Case Study: Demand Forecast in Woolworths





There are two main approaches:

Quantitative	Uses historical data and statistical models (like regression analysis)
Qualitative	Relies on expert opinions, market research, or customer feedback

A key method for demand forecasting involves the use of regression models, which help identify and quantify the impact of various factors on future demand.

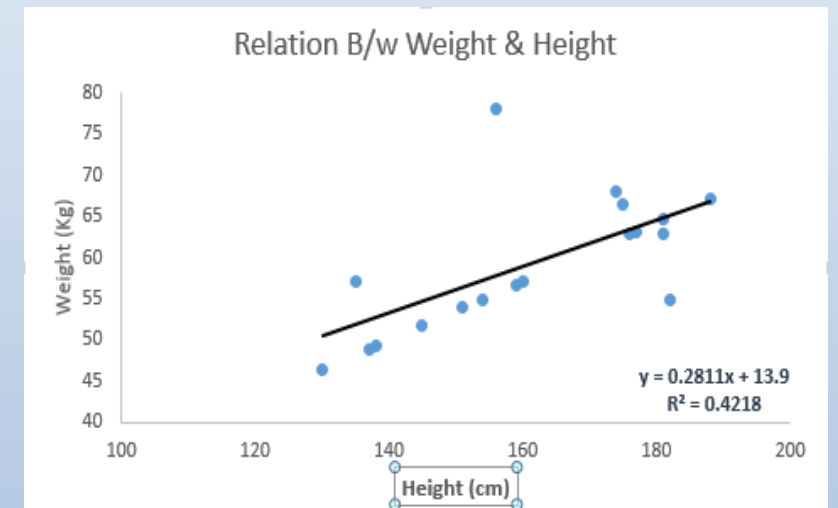


Regression

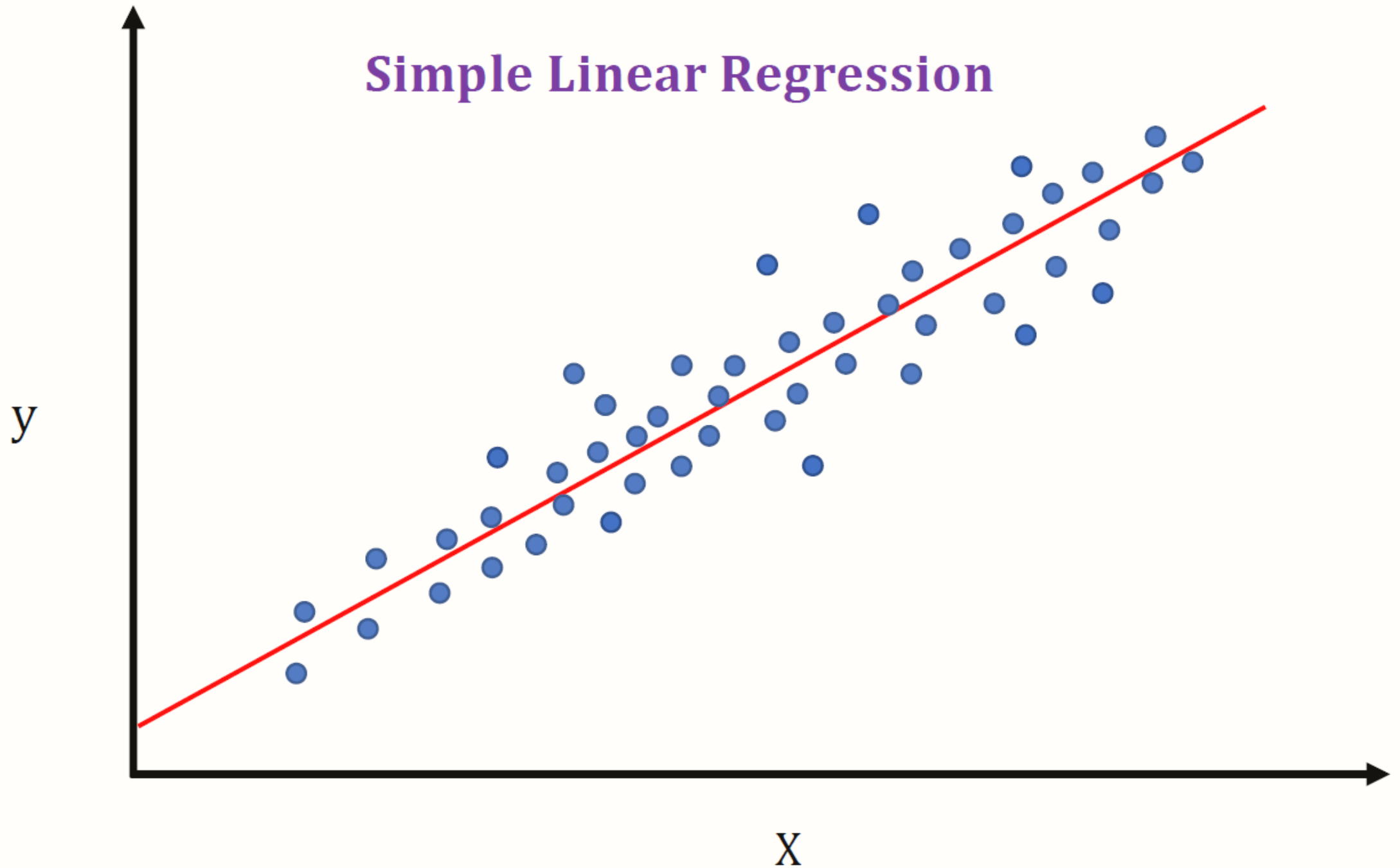
- Regression investigates the relationship between a **dependent (target)** variable and **independent variables (predictor)**
- It is used for forecasting, time series analysis and finding causal effect relationship between variables

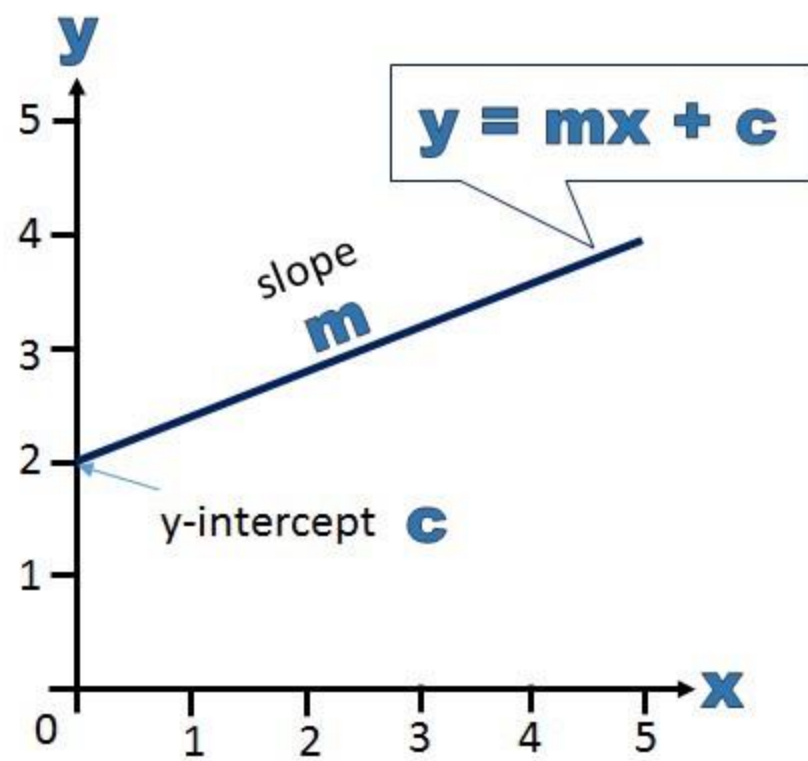
Linear Regression

- Establishes a **relationship between dependent variable (y) and one or more independent variables (X)** using best fit straight line (also known as regression line)
- This is represented by an **equation $y = a*x + b$** , where a is the intercept, b is the slope of the line and e is the error term
- The **available data is graphed to find this regression line. This enables determining future values.** The equation can then be found to predict the value of the target variable based on the independent variables

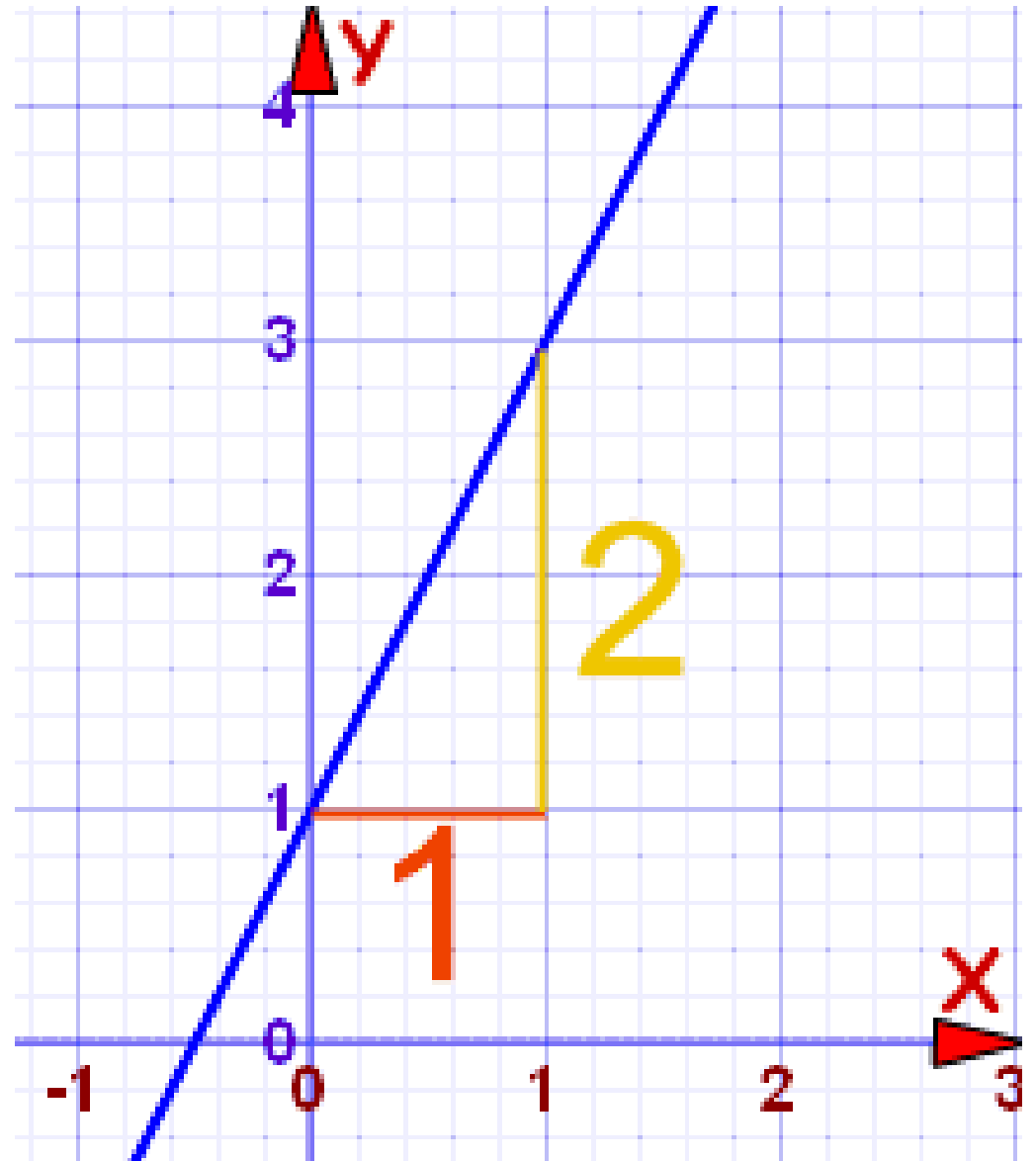


Simple Linear Regression





$$Y = ?x + ?$$



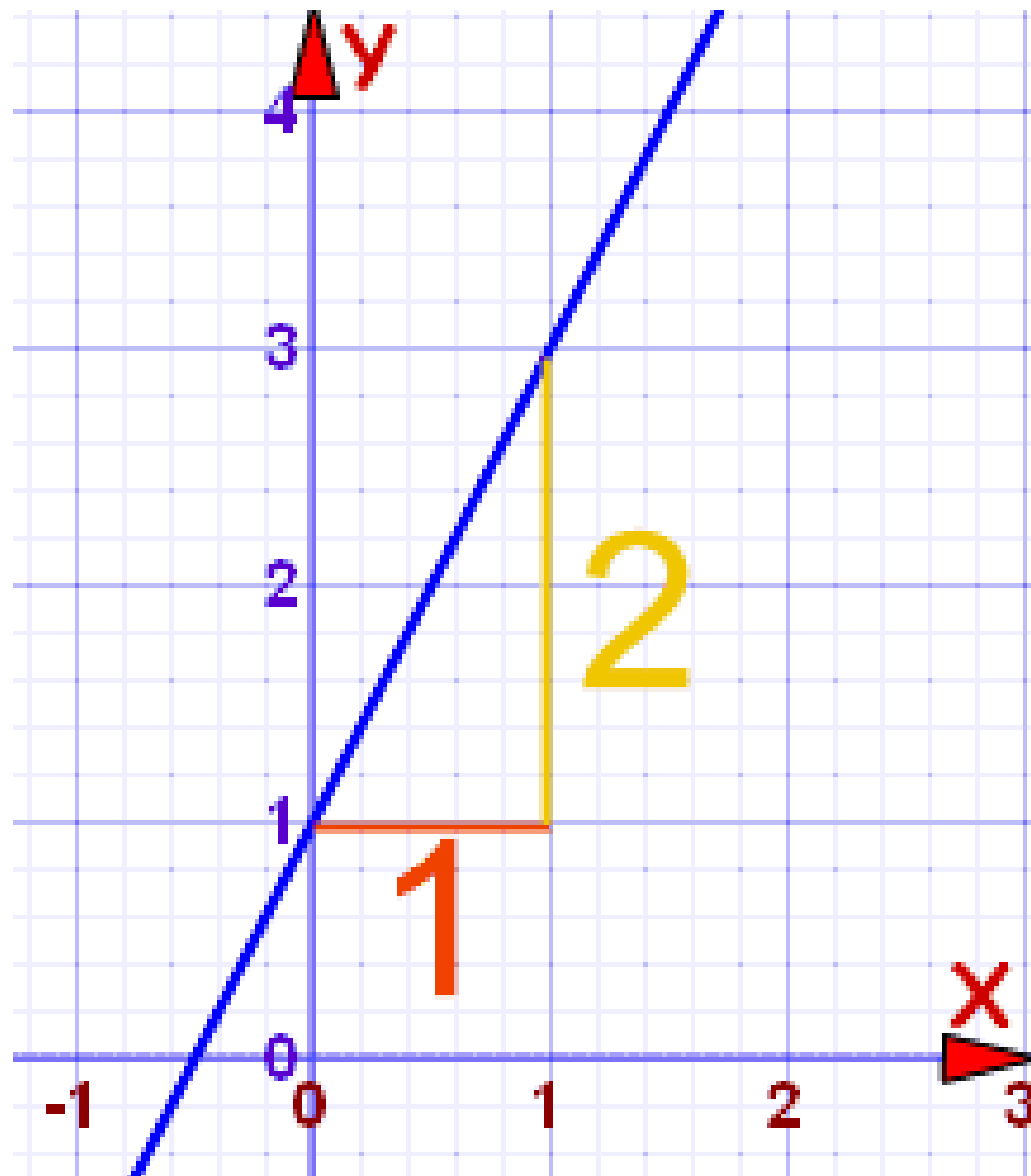
$$Y = ?x + ?$$



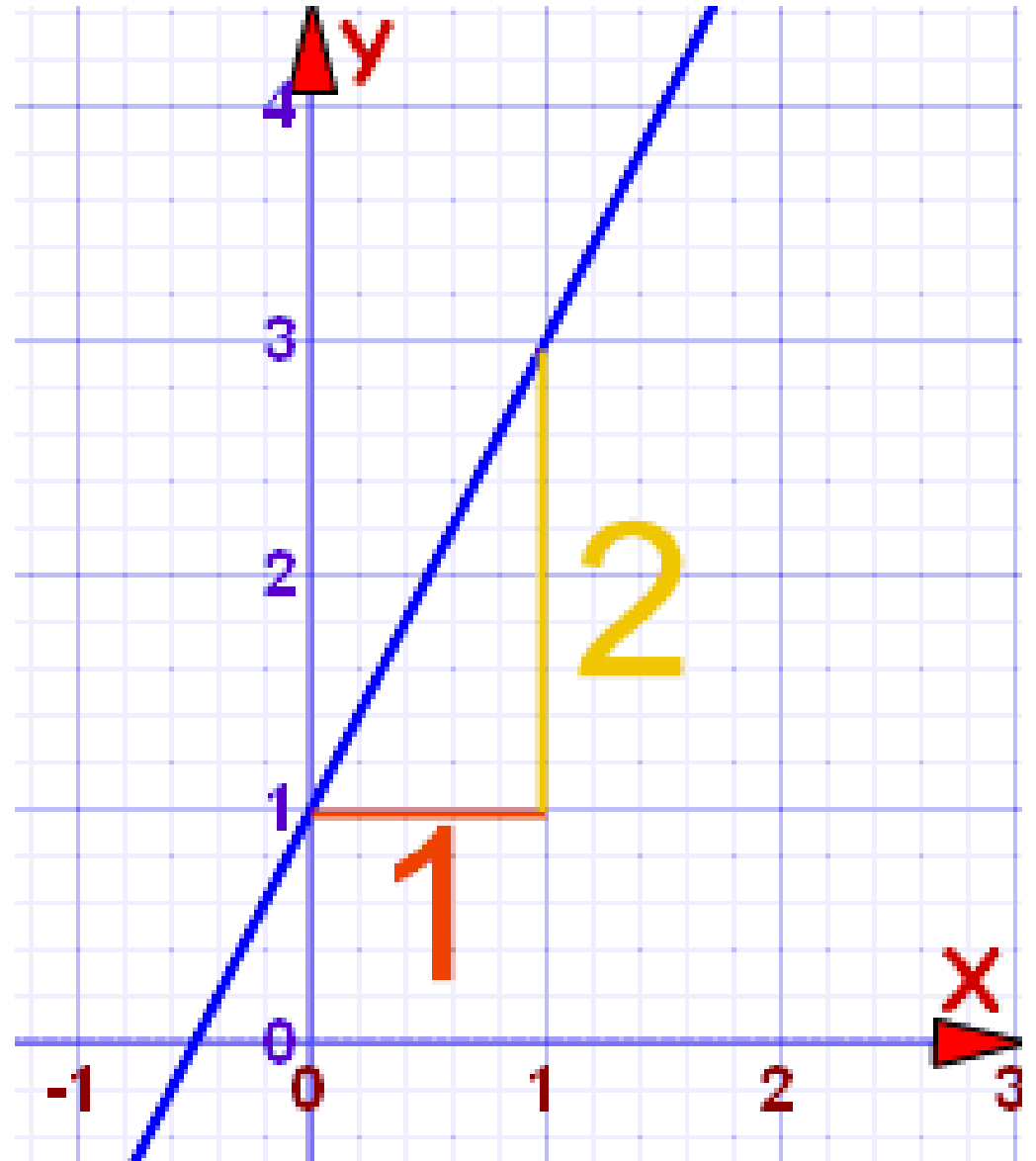
dep.



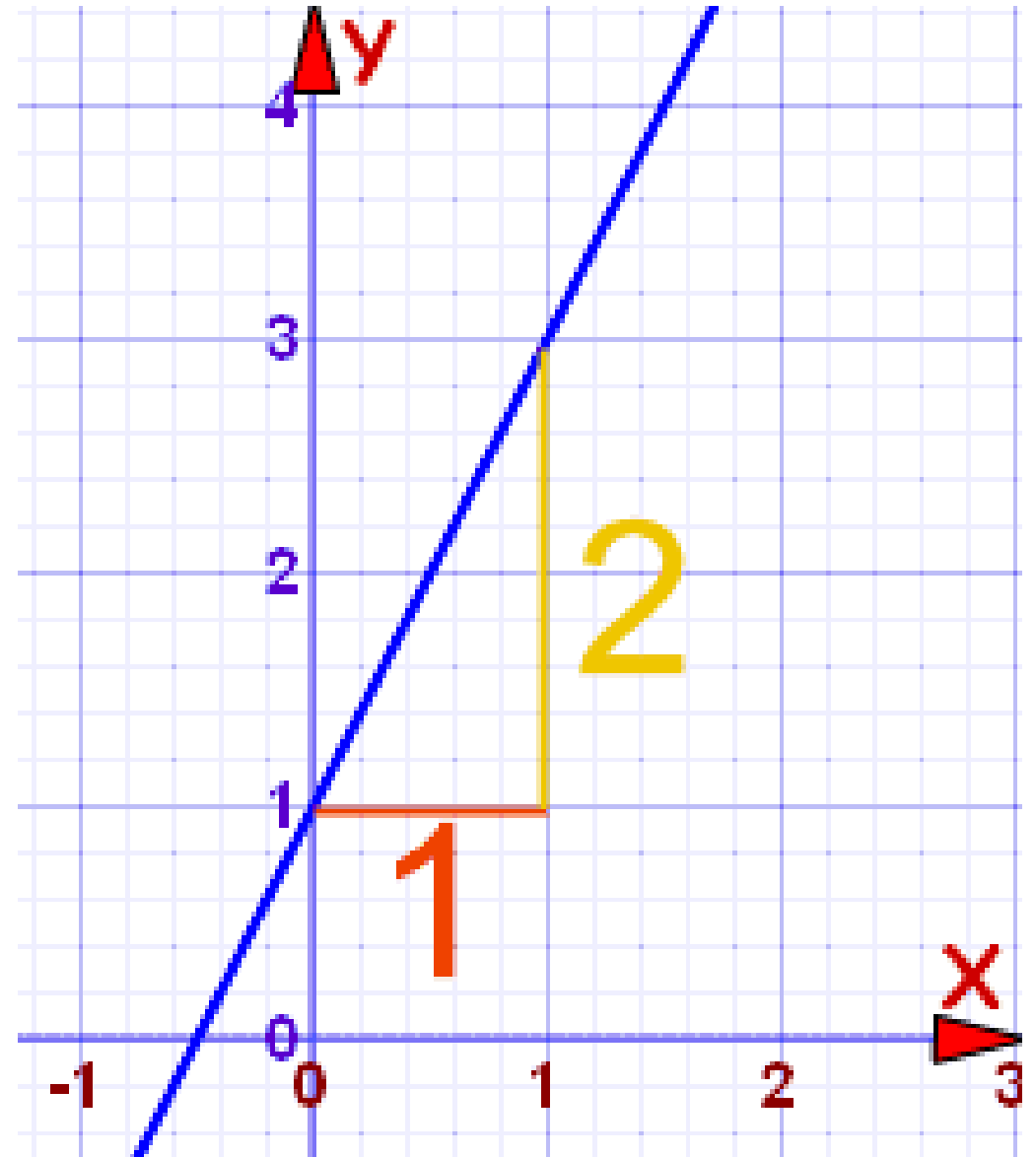
Ind.



$$Y = ax + b$$



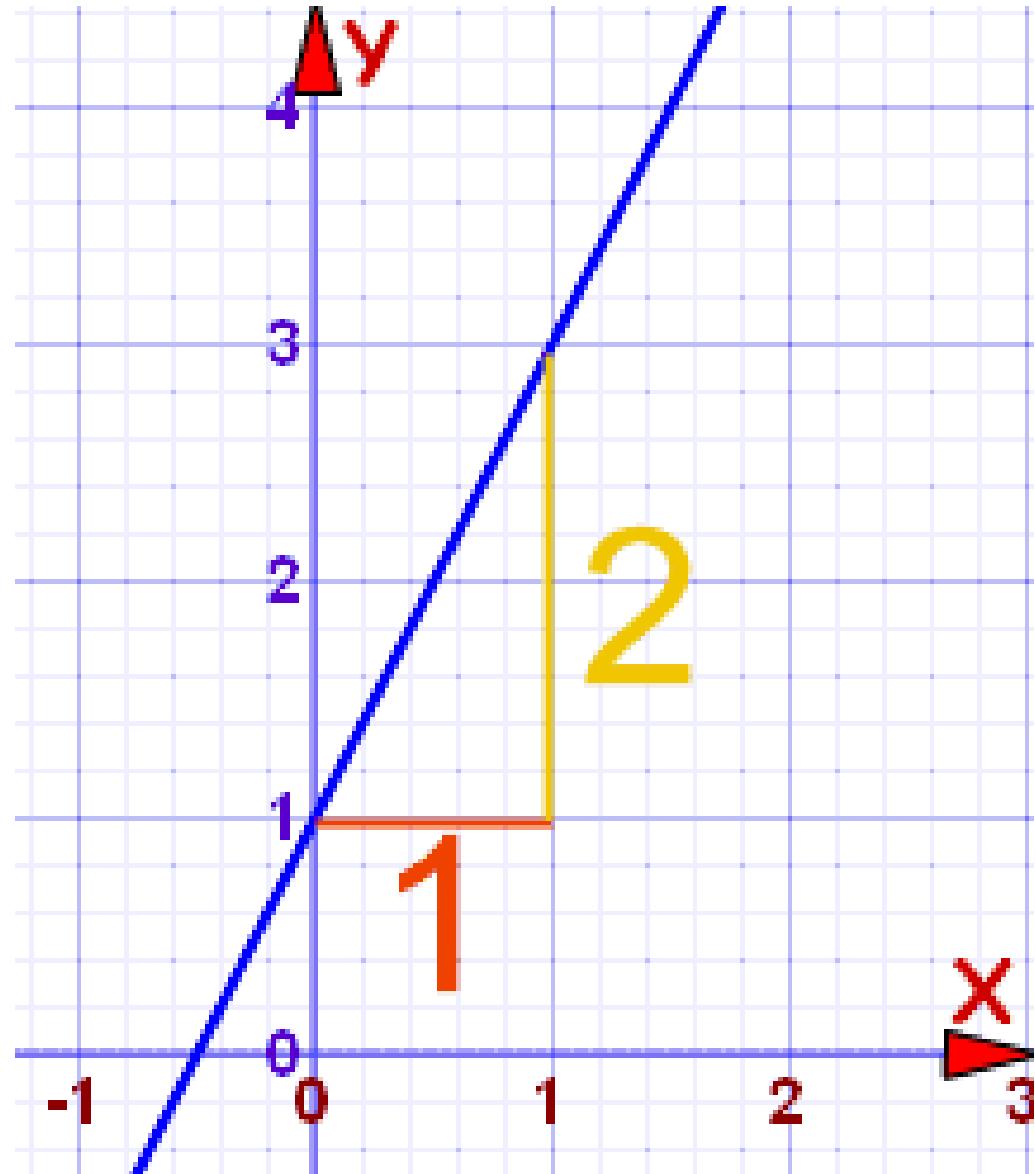
$$Y=2x+1$$



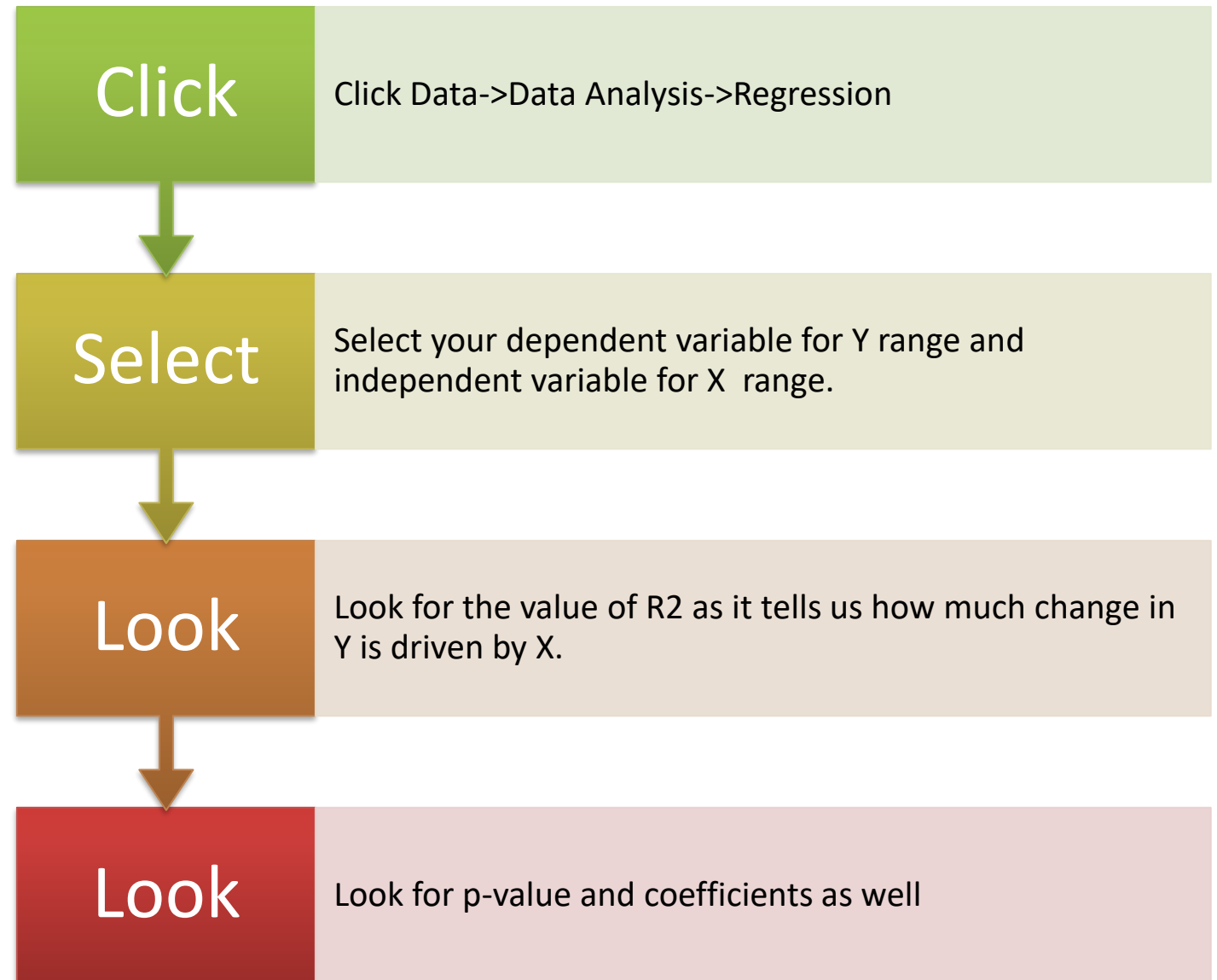
$$Y=2x+1$$

$$Y=mx+b$$

$$Y=ax+b$$



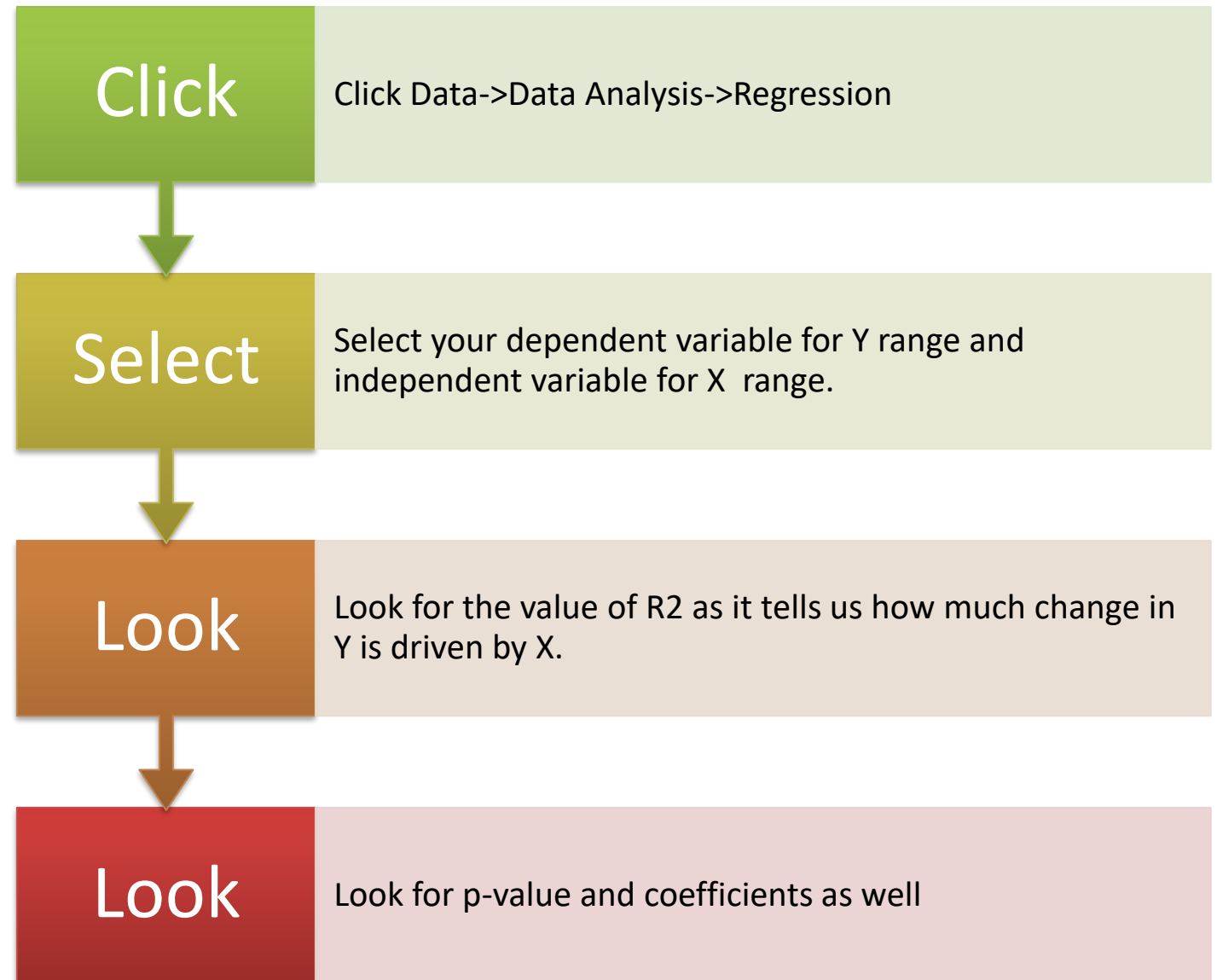
Example 1: The relationship between experience (years with company) and salary





An Example of Linear Regression

Example 1: The relationship between experience (years with company) and salary

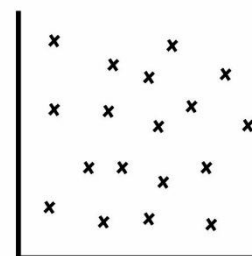


SUMMARY OUTPUT								
Regression Statistics								
Multiple R	0.766051647							
R Square	0.586835127							
Adjusted R Square	0.583883949							
Standard Error	15568.50444							
Observations	142							
ANOVA								
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>			
Regression	1	48196392870	48196392870	198.8477796	1.17585E-28			
Residual	140	33932966285	242378330.6					
Total	141	82129359155						
	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	69816.58422	3023.145185	23.09402293	1.35128E-49	63839.66379	75793.50466	63839.66379	75793.50466
Year with Company	12185.36839	864.1284236	14.10133964	1.17585E-28	10476.94008	13893.79671	10476.94008	13893.79671

SUMMARY OUTPUT

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Adjusted R Square	0.583883949
Standard Error	15568.50444
Observations	142

Positive
CorrelationNegative
CorrelationNo
Correlation

ANOVA

	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	48196392870	48196392870	198.8477796	1.17585E-28
Residual	140	33932966285	242378330.6		
Total	141	82129359155			

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Year with Company	12185.36839	864.1284236	14.10133964	1.17585E-28	10476.94008	13893.79671	10476.94008	13893.79671

R-squared values determines the proportion of variance in the dependent variable that can be explained by the independent variable. It range from 0 to 1 and are commonly stated as percentages from 0% to 100%.

A R-squared between 0.50 to 0.99 is acceptable

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.766051647
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Observations	142

ANOVA

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ANOVA

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Regression	1	48196392870	48196392870	198.8477796	1.17585E-28
Residual	140	33932966285	242378330.6		
Total	141	82129359155			

Similar to p-Value in diagnostic analytics, here, in regression, Significance F needs to be below 0.05 to proceed



Multiple Regression

- This is determining the relationship between **multiple independent variables** and a **dependent variable**. The dependent variable is modelled as a function of several independent variables with corresponding coefficients, along with the constant term.
- Multiple regression requires **two or more independent variables** which is why it's called multiple regression.
- It can be represented by:

$$Y = a_1x_1 + a_2x_2 + \dots + a_nx_n + b$$

Assuming that your main variables to determine the price of your products are the production cost and inventory cost. Find the regression formula for the existing products.

This will enable you to **predict/determine** what should be the price of your newly developed product (or generally, **future products**)





Write the price sale formula

The dependent variable is the price sale

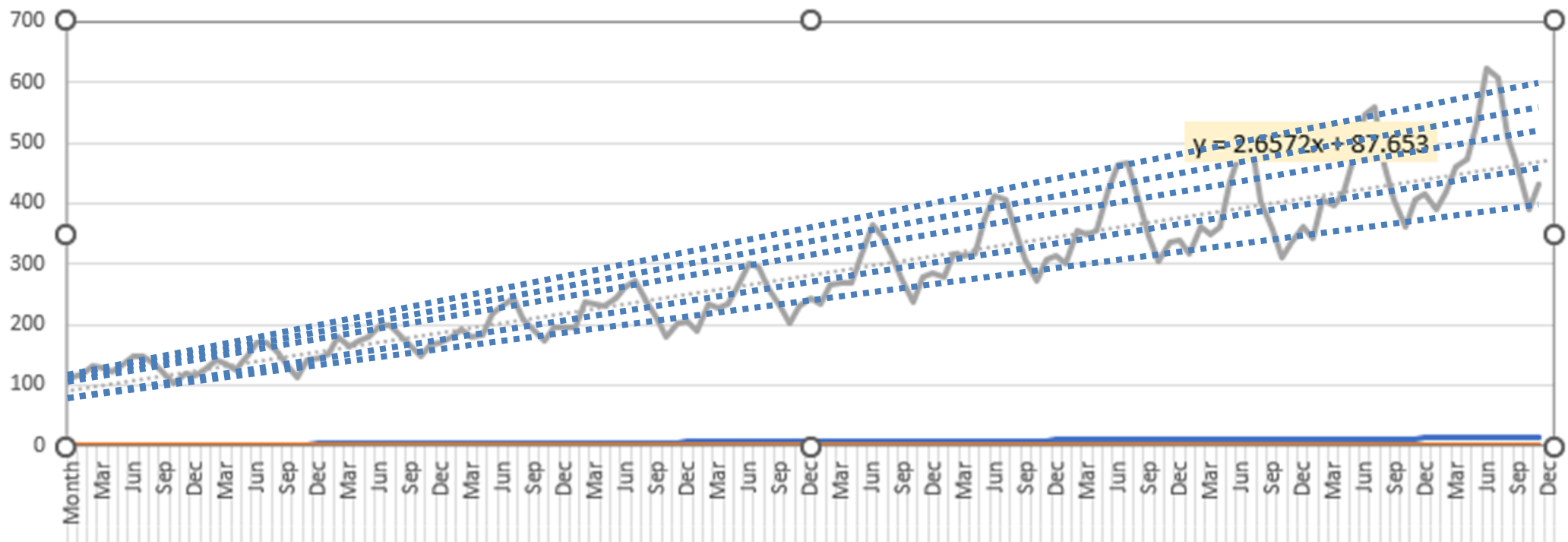
The dependent variables are manufacturing and inventory costs

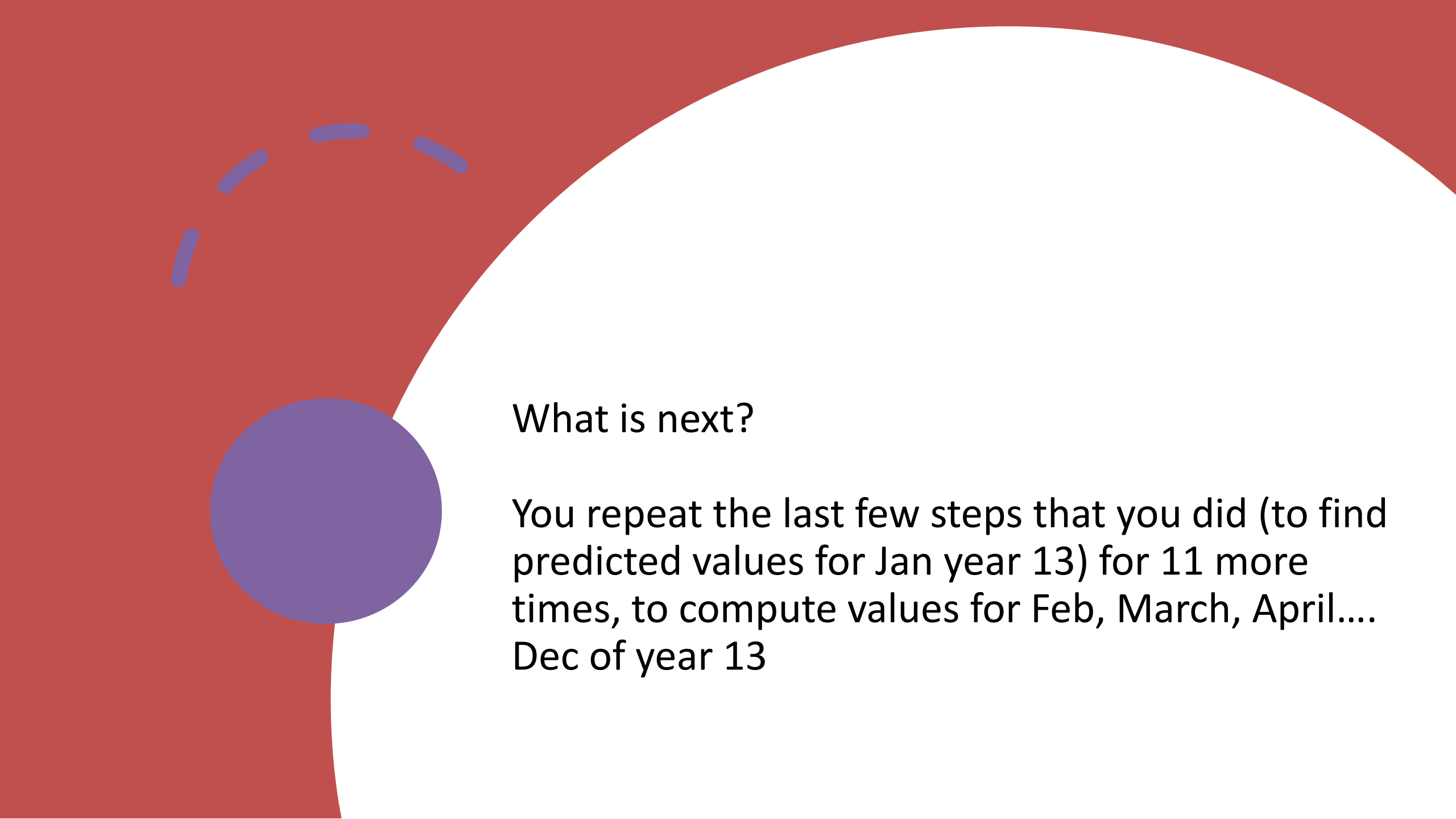
The rest are fixed costs (your intercept value)

Which one has a higher impact on product price, production cost or inventory cost?

Product price = a * Production Cost + b * Inventory cost + c		

#Passengers in 12 years





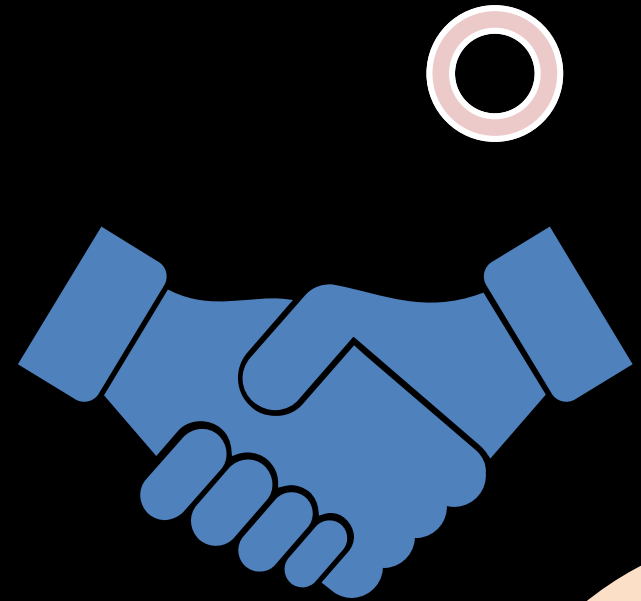
What is next?

You repeat the last few steps that you did (to find predicted values for Jan year 13) for 11 more times, to compute values for Feb, March, April....
Dec of year 13



What about non-linear
models???

A key problem in
Demand Forecast:
Information Holding
and Trust Issue



*Biggest Problem is **TRUST***

Who should optimise the supply chain and why?

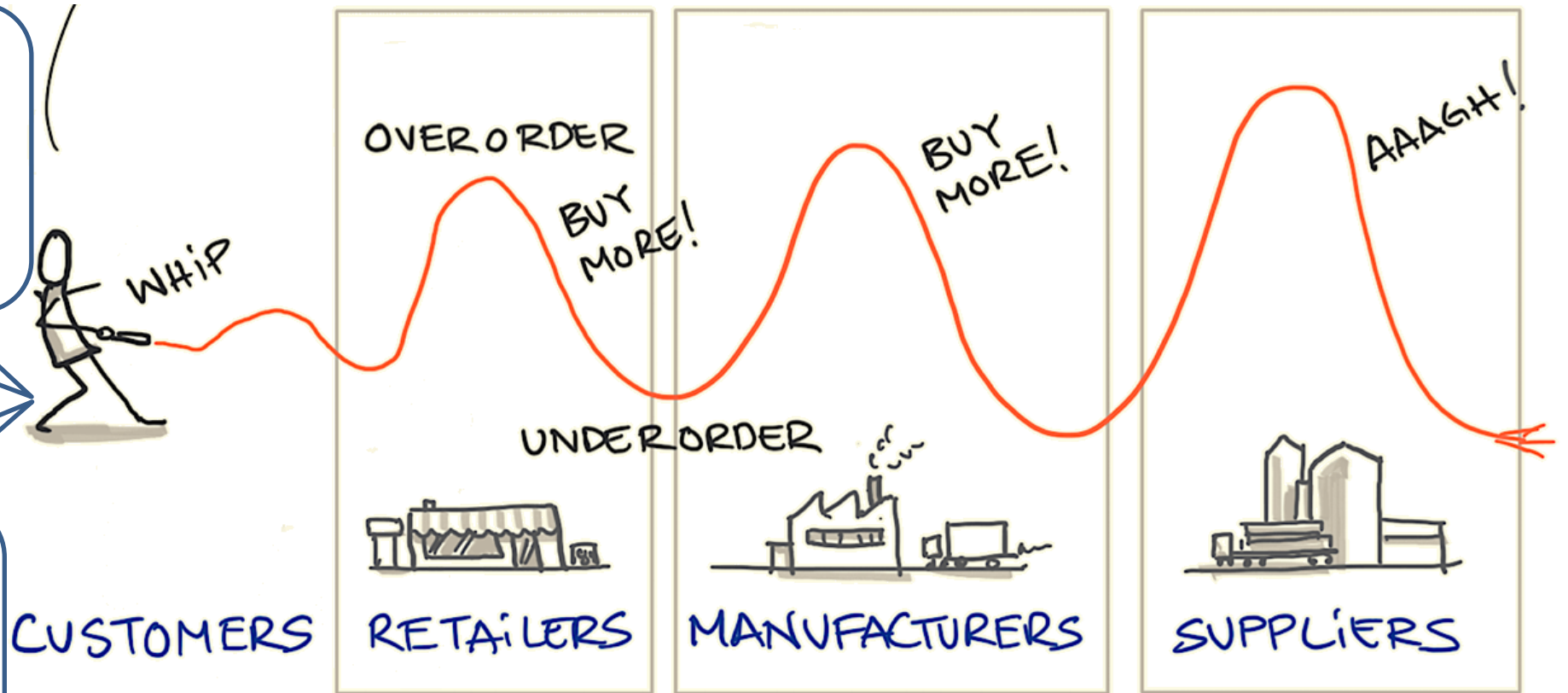
How the benefits will be shared?

Is it fair? Who is taking advantage?

The Bullwhip Effect

1. Access to real demand information?
2. Everyday low price
3. Collaborative forecasting
4. Collaborative planning

1. Demand forecasting
2. Lead time
3. Batch ordering
4. Price fluctuations
5. Inflated orders



sketchplanations

The Bullwhip Effect: Causes

- **1. Consumer Demand Fluctuates Slightly**

- A small uptick in retail sales — say, an increase of 5% more sale this week — triggers concern about future demand.
- Retailers interpret this as a trend and increase their next order by 10% to avoid stockouts.

- **2. Retailer Sends Amplified Orders to Wholesaler**

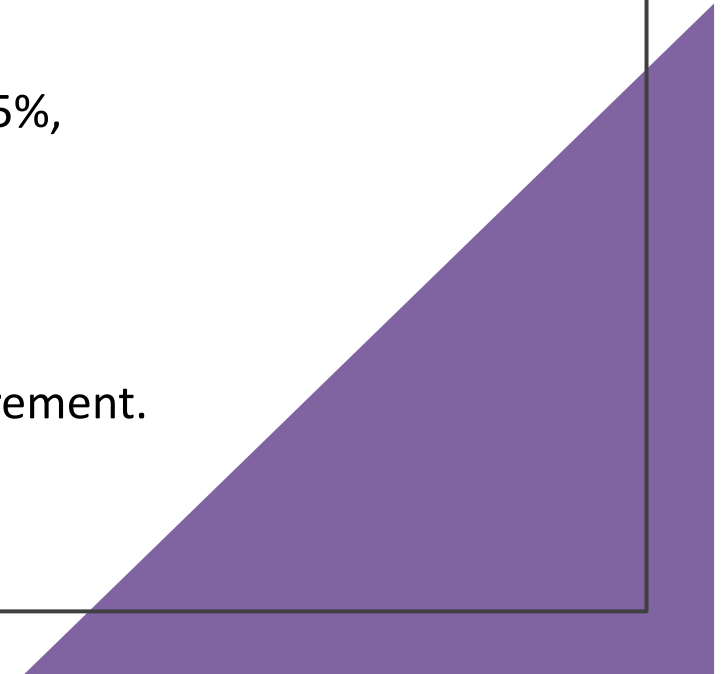
- The wholesaler sees a 10% increase in orders and assumes demand is rising.
- To buffer against uncertainty and lead times, they order 15% more from the manufacturer.

- **3. Manufacturer Overreacts**

- The manufacturer sees a 15% spike and ramps up production by 20–25%, fearing delays or missed opportunities.
- They may also increase raw material purchases, hire more staff, or run overtime shifts.

- **4. Supplier Responds to Inflated Signals**

- Raw material suppliers see the surge and scale up extraction or procurement.
- This leads to overproduction, excess inventory, and potential waste — especially if the original demand spike was temporary or seasonal.



The Bullwhip Effect: Strategies to Eliminate

- **Real-Time Information Sharing** Use integrated systems (e.g., ERP, EDI) to share POS data, inventory levels, and forecasts across the chain.
- **Collaborative Planning** Adopt frameworks like **CPFR (Collaborative Planning, Forecasting, and Replenishment)** to align decisions.
- **Vendor Managed Inventory (VMI)** Let suppliers manage inventory based on actual consumption, not periodic orders.
- **Smaller, Frequent Orders** Reduce batching to smooth demand signals.
- **Stable Pricing Strategies** Avoid erratic promotions that distort demand.
- **Lead Time Reduction** Streamline logistics and production to reduce uncertainty.
- **Advanced Analytics & AI Forecasting** Use predictive models that incorporate external factors (weather, trends, events) for smarter planning.



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Long-Term Relationship with Suppliers and Distributors

